

static electric field E produced by charges.

Always from + to - . diverges of converged

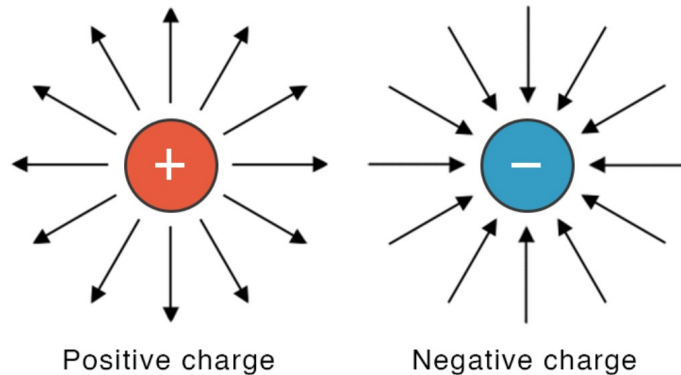
Voltage drop is to charge what height drop is to mass.

(E wants to lower voltage of a positive charge. Gravity wants to lower height. $\rightarrow$ lower the PE of the system)

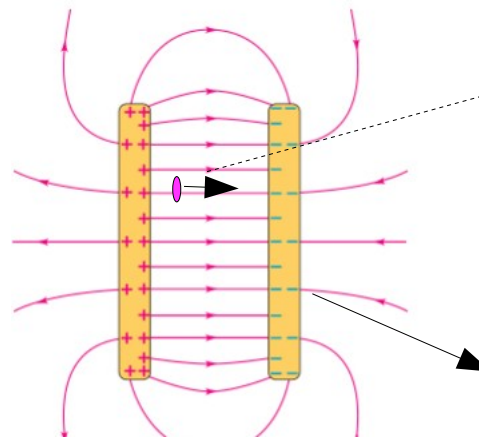
A distribution of charge creates an electric field E in the space around it. A charge q in the electric field “feels” the field and speeds up. It is acted up by an electric force. $F_e = q E$. A positive charge moves from a high potential to a low potential = high voltage to low voltage.

A positive charge goes along the electric field.

A negative charge wants to go from low potential to high potential. Against the electric field.



A charge placed inside
Will accelerate. Example: electrons in a cathode tube
Speed up from - to +.



The electric field is a vector field.

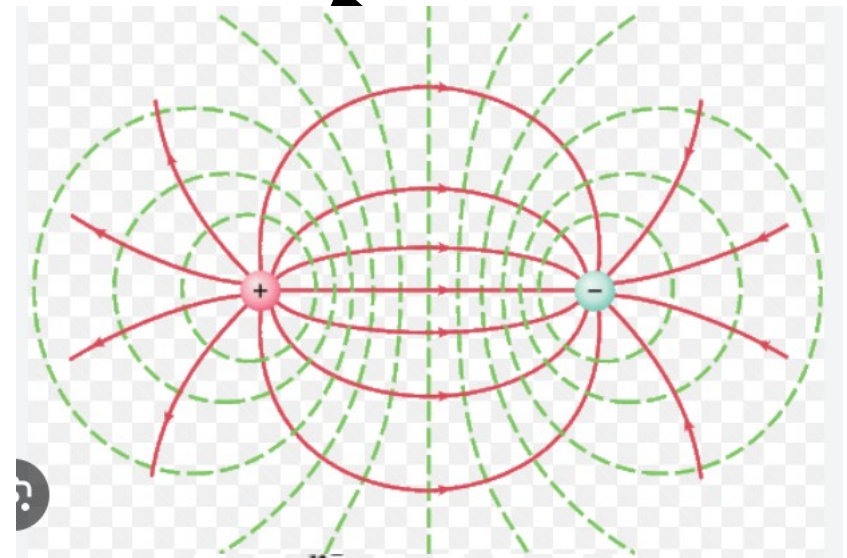
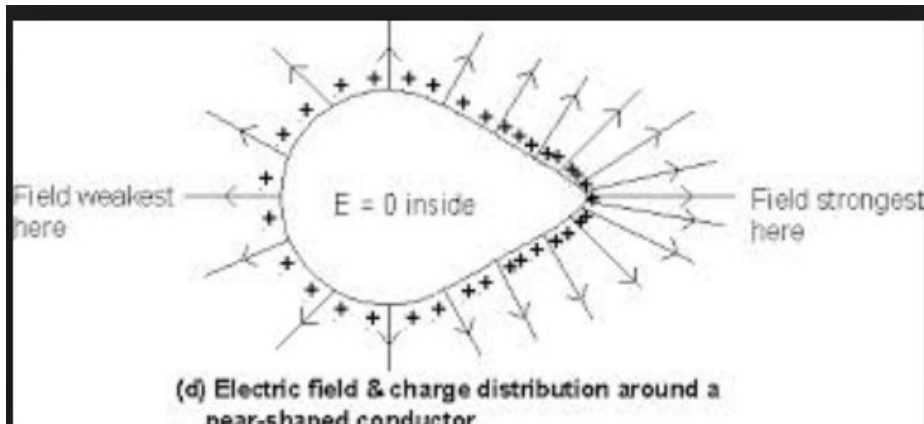
A positive charge placed
Inside an electric field will accelerate.
Like a mass in a gravitational field

capacitors



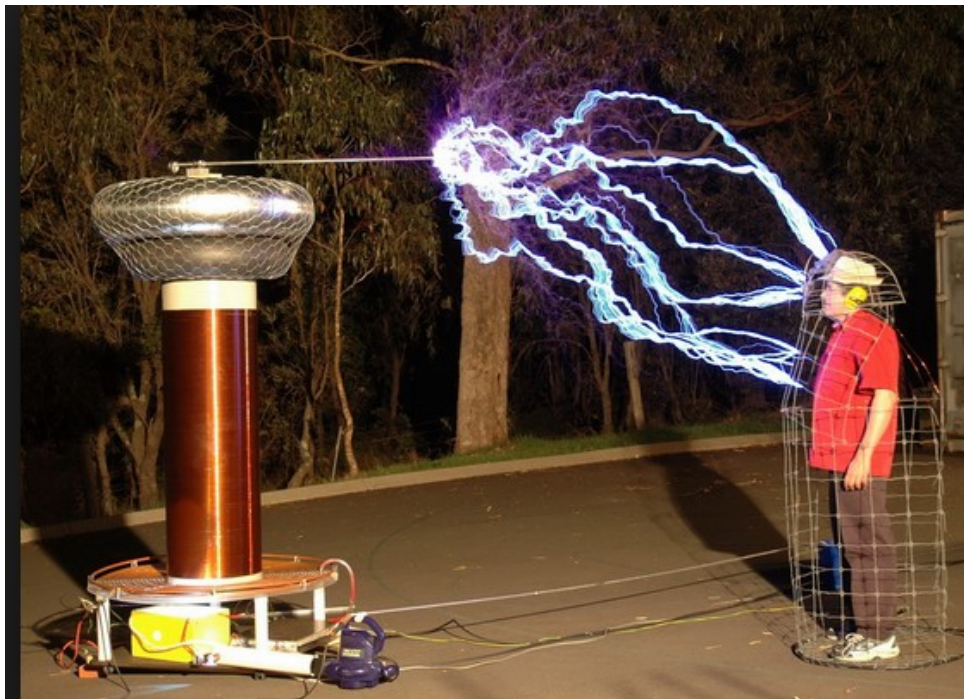
All appliances plugged in
Produce electric field

Electric field is perpendicular to equipotentials. The surface and inside of A conductor is an equipotential.



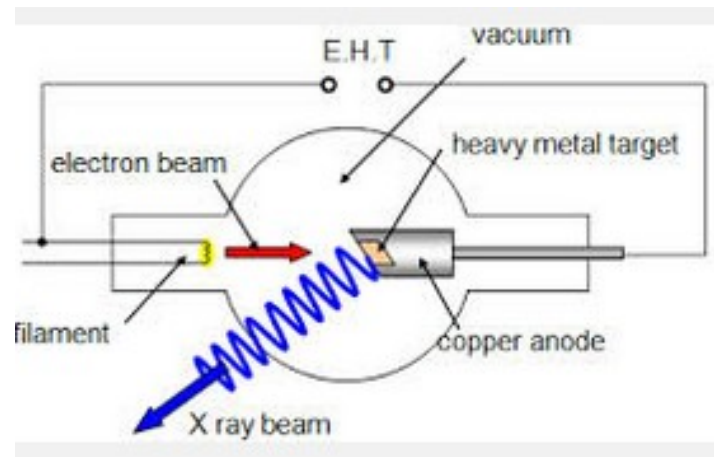
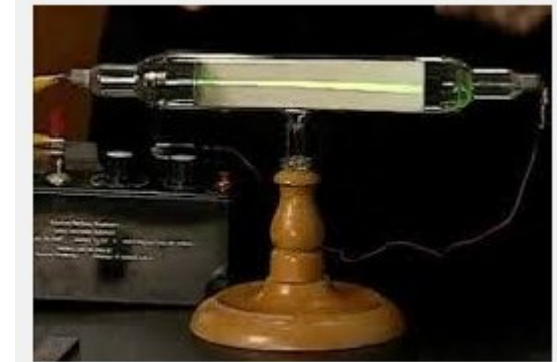
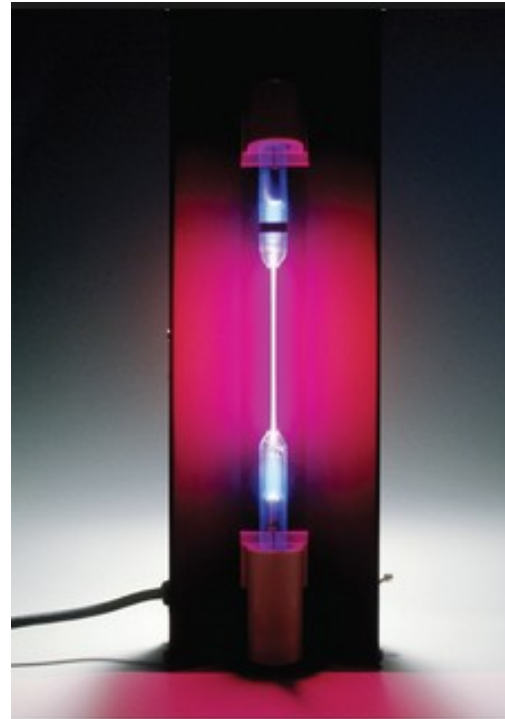
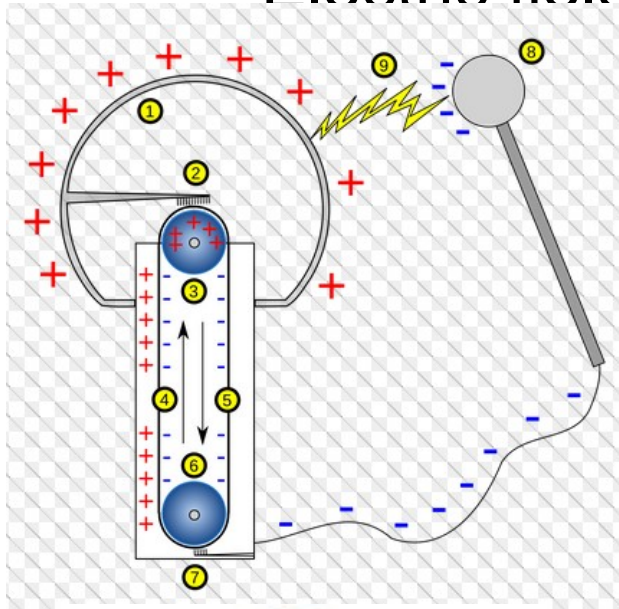
where $\vec{p} = q\vec{d}$ is defined as the dipole moment.

Electric dipole. ,p is the electric dipole moment. Its a vector From - to +

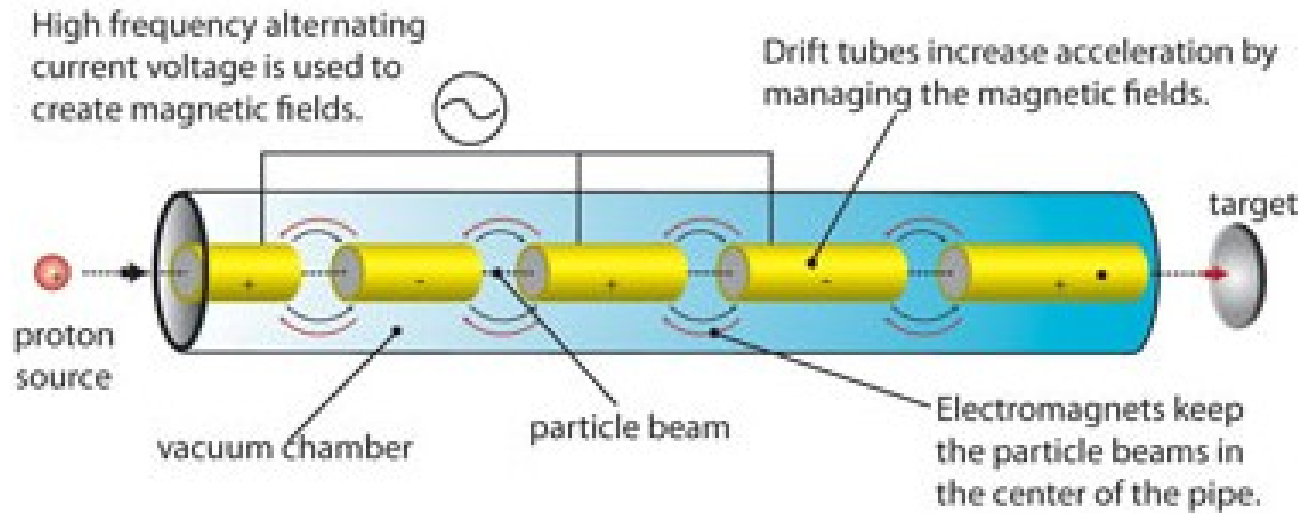


Faraday's cage.
Charges stay on the outside
Of a conductor.
Inside a conductor $E=0$, $V=cste$
External E can not penetrate
A conductor.

Electric field between high voltage and ground



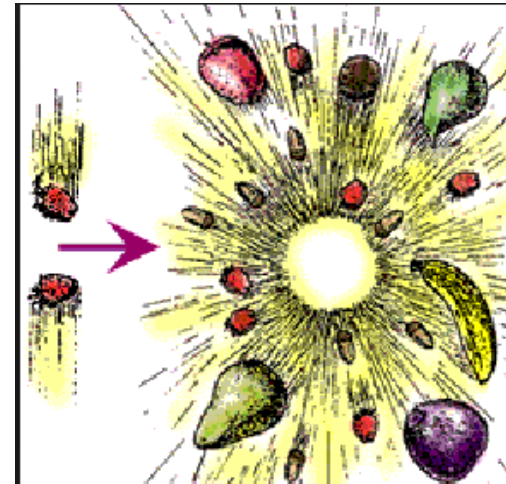
Linear Accelerator



Watch movie about
accelerator

<http://atomic.lindahall.org/what-is-an-atom-smasher.html>

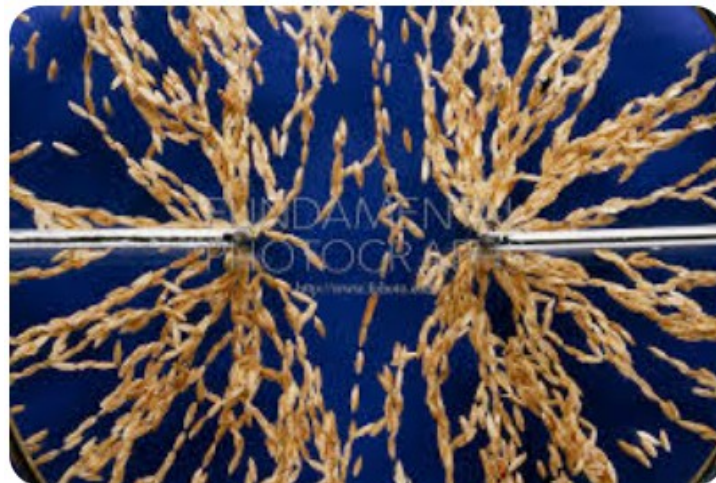
<http://www.triumf.ca/isotope-production>



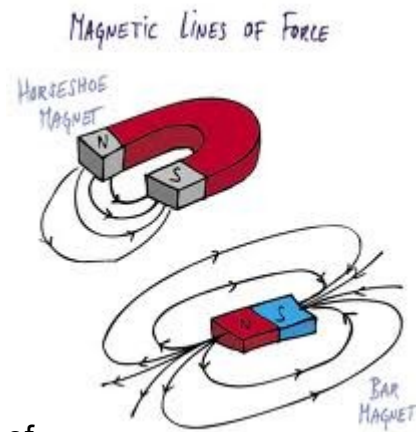
Vacuum permittivity,

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2$$

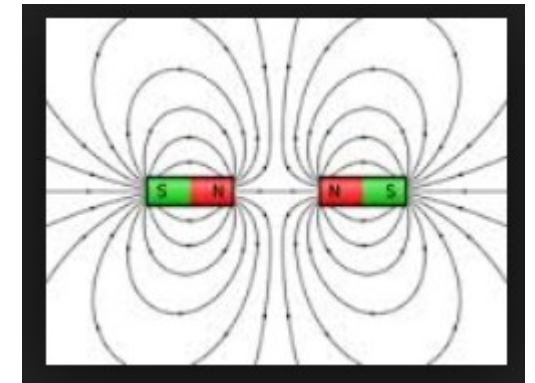
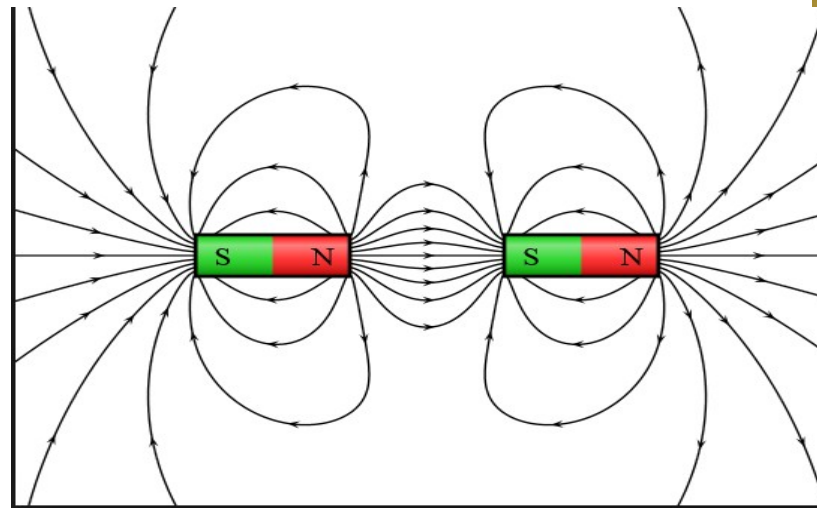
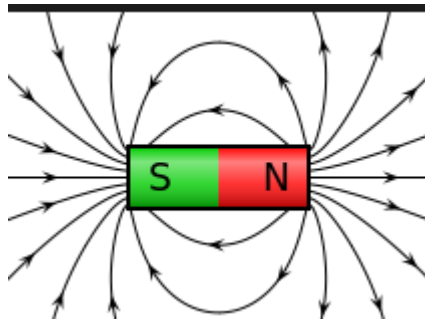
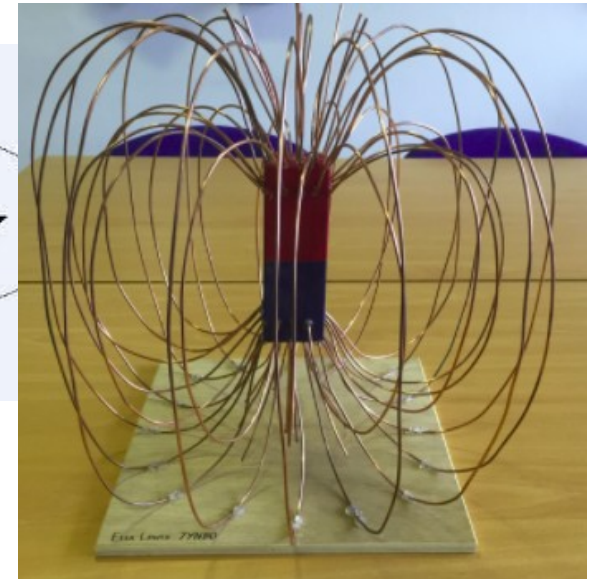
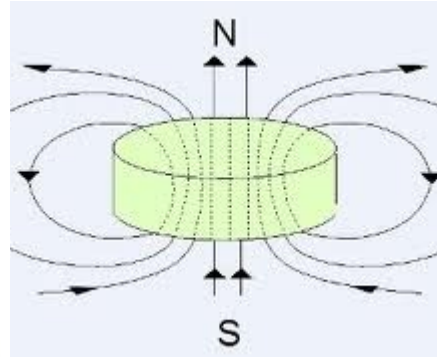
Coulomb's Law (Electrostatic Force)	$F_e = \frac{kQq}{r^2}$
Electric Field	$E = \frac{F_e}{q} = \frac{kQ}{r^2} = \frac{V}{d}$
Electric Potential Energy	$U = \frac{kQq}{r}$
Electric Potential	$V = \frac{U}{q} = \frac{kQ}{r}$



2) static magnetic field

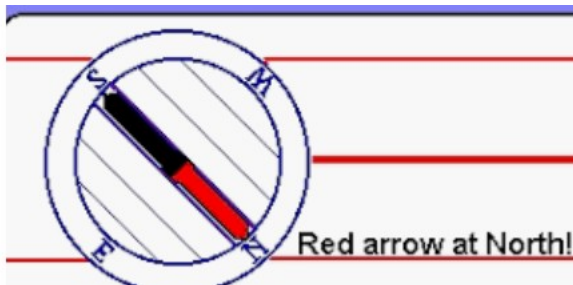
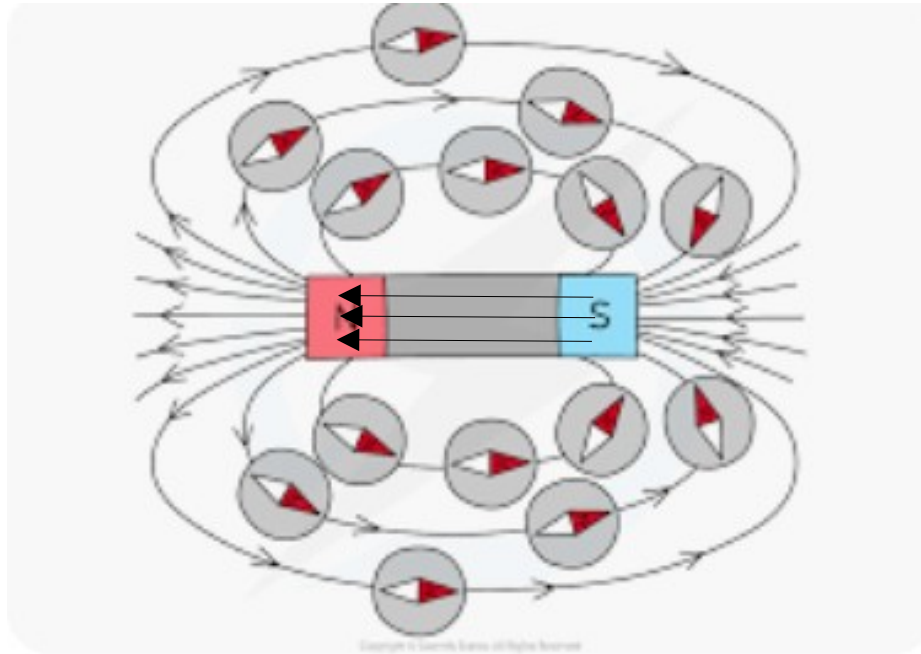


We can use iron filing
To map the field. The little pieces of
Iron get magnetized and behave like
Tiny compasses.

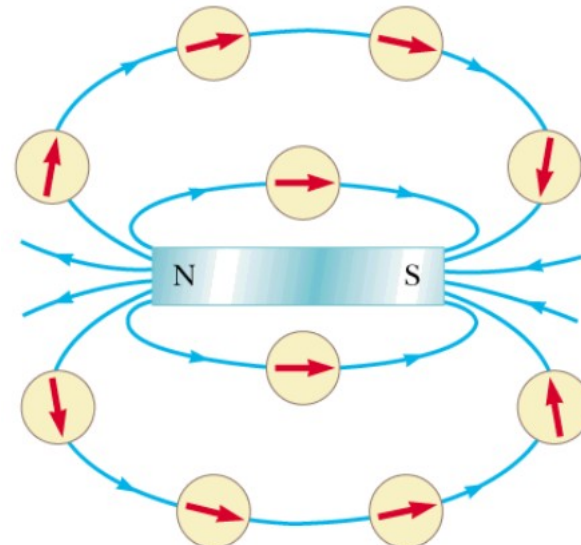


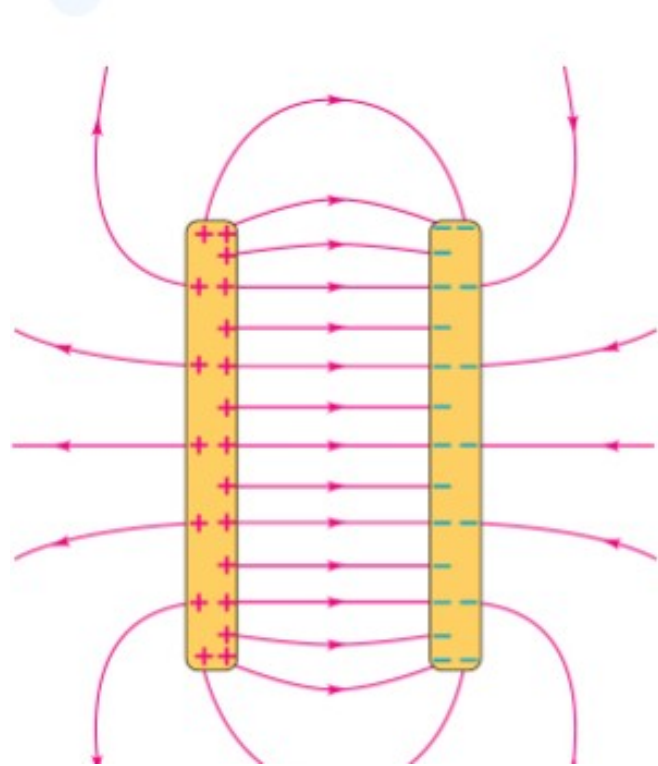
Show that the configuration of the magnetic field is like the electric field of a dipole.

We can use iron filing to map the field. The little pieces of Iron get Magnetized and behave like Tiny compasses.

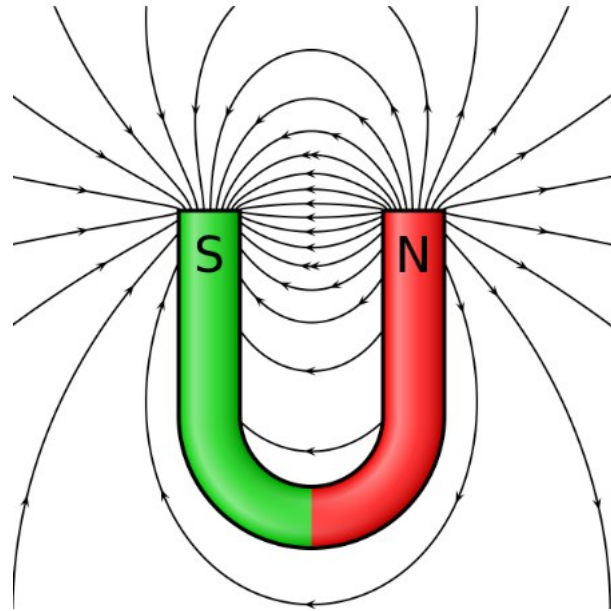


A bar magnet like inside a compass. Is showing you the vector B , magnetic field at a given Point.

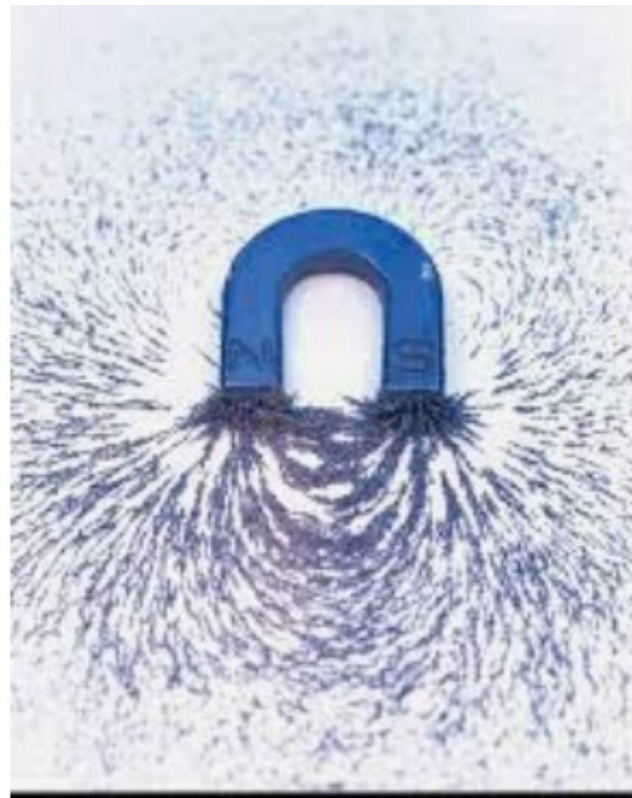




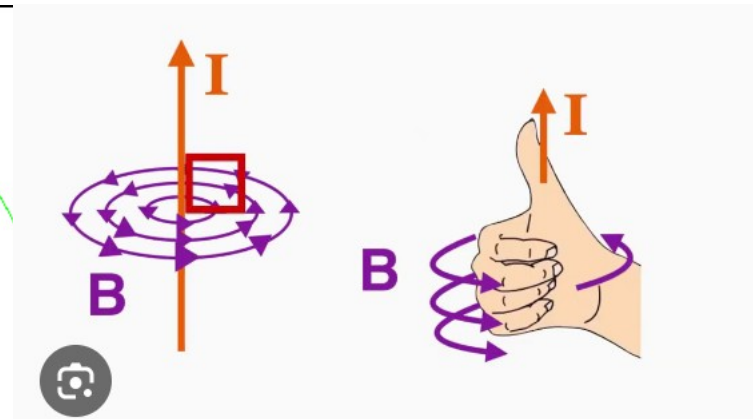
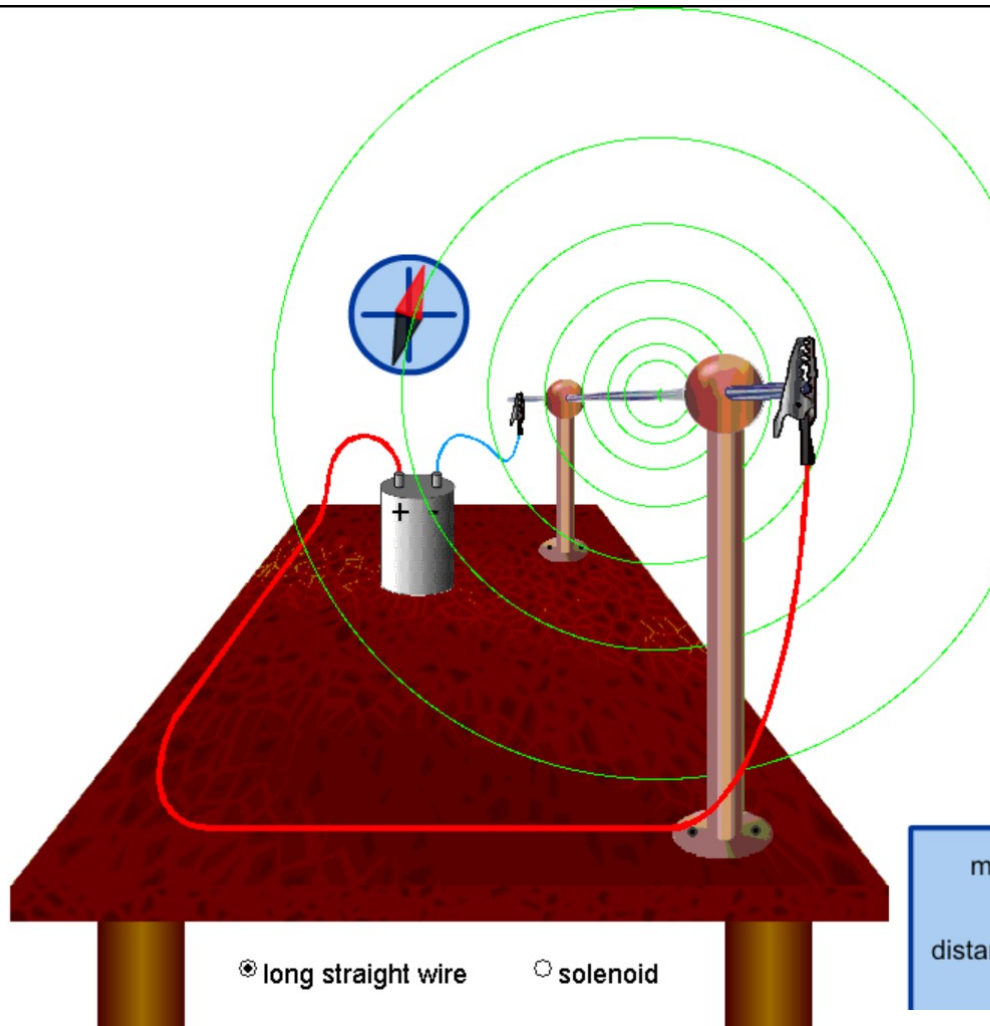
Uniform electric field
Seeds in oil



Uniform magnetic field:
Iron filing sprinkled



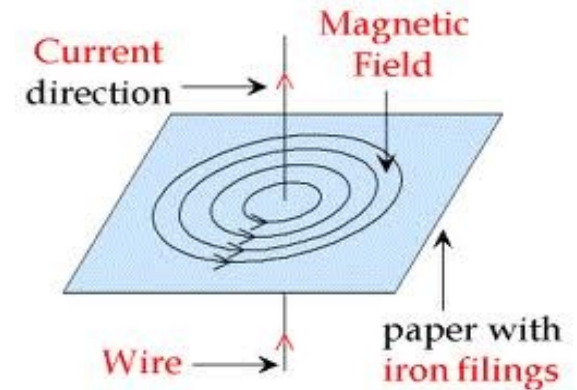
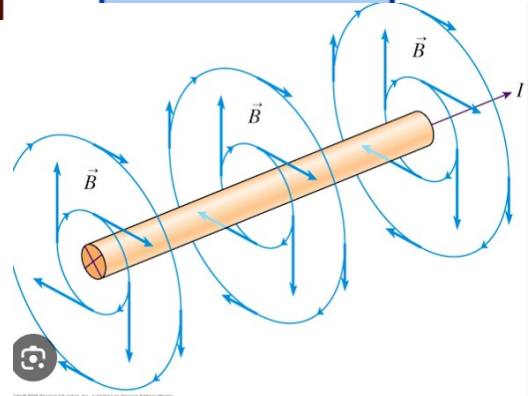
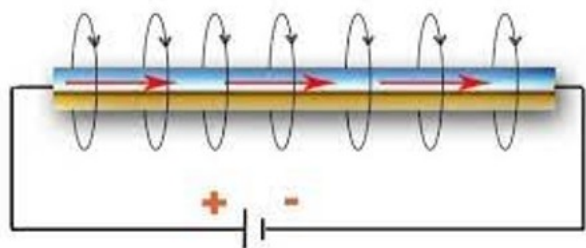
1820 Oersted experiment: the source of magnetism is electricity

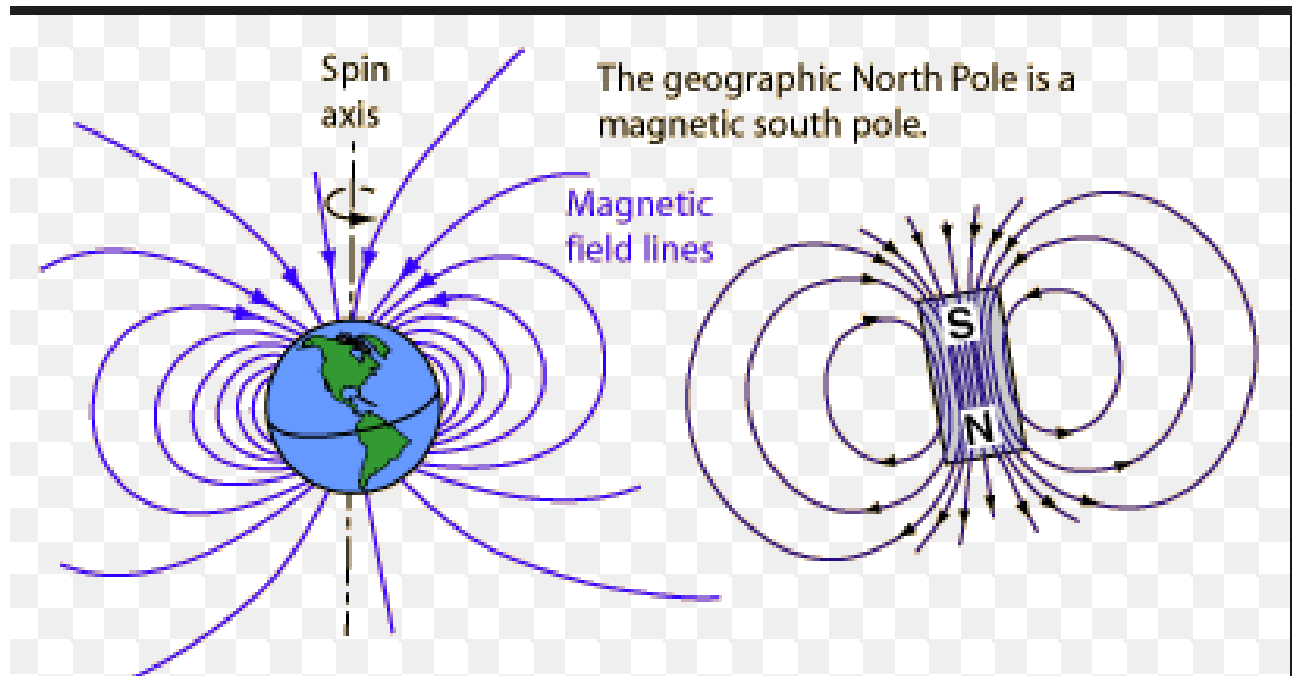


...the right-hand rule which
Way B is flowing.
The magnetic is a vector.

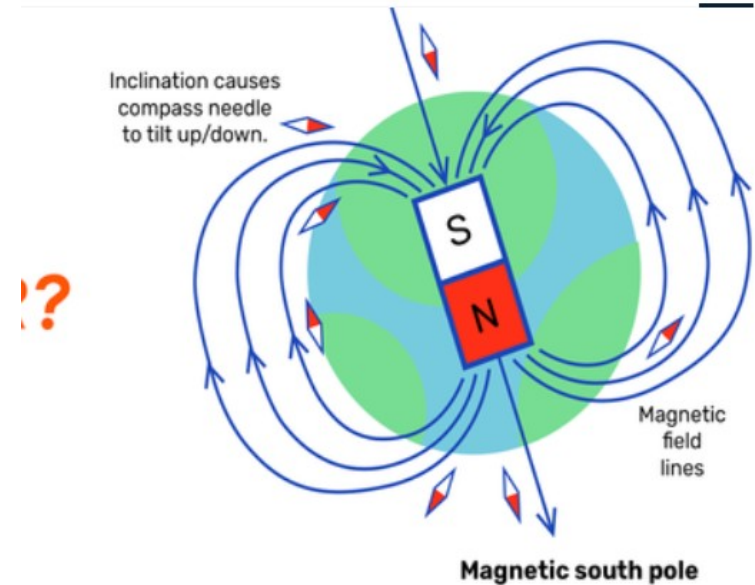
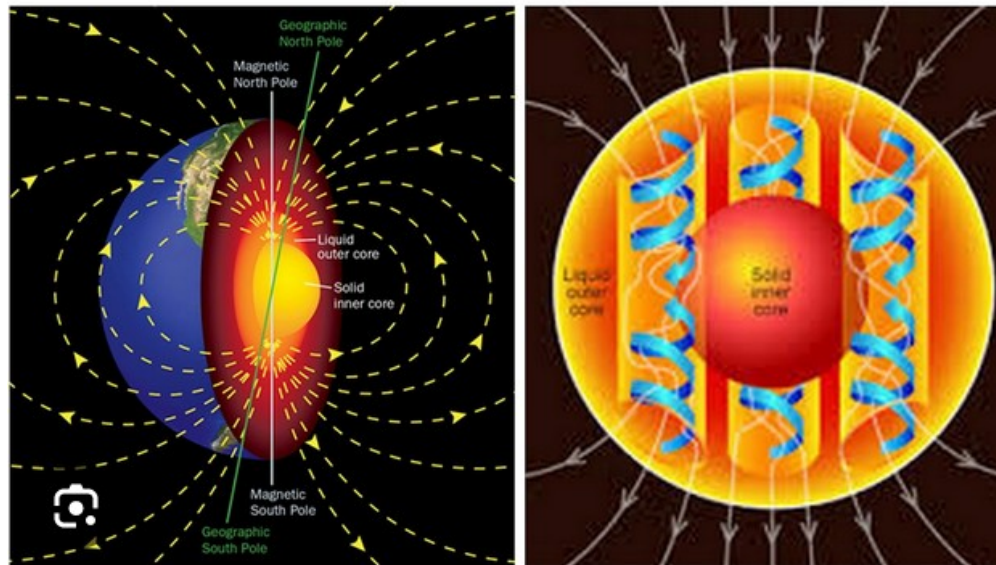
The magnitude is inversely
Proportional to the distance
From wire. (see contour lines)

magnetic field:
0.046 T
distance from center:
0.173 m

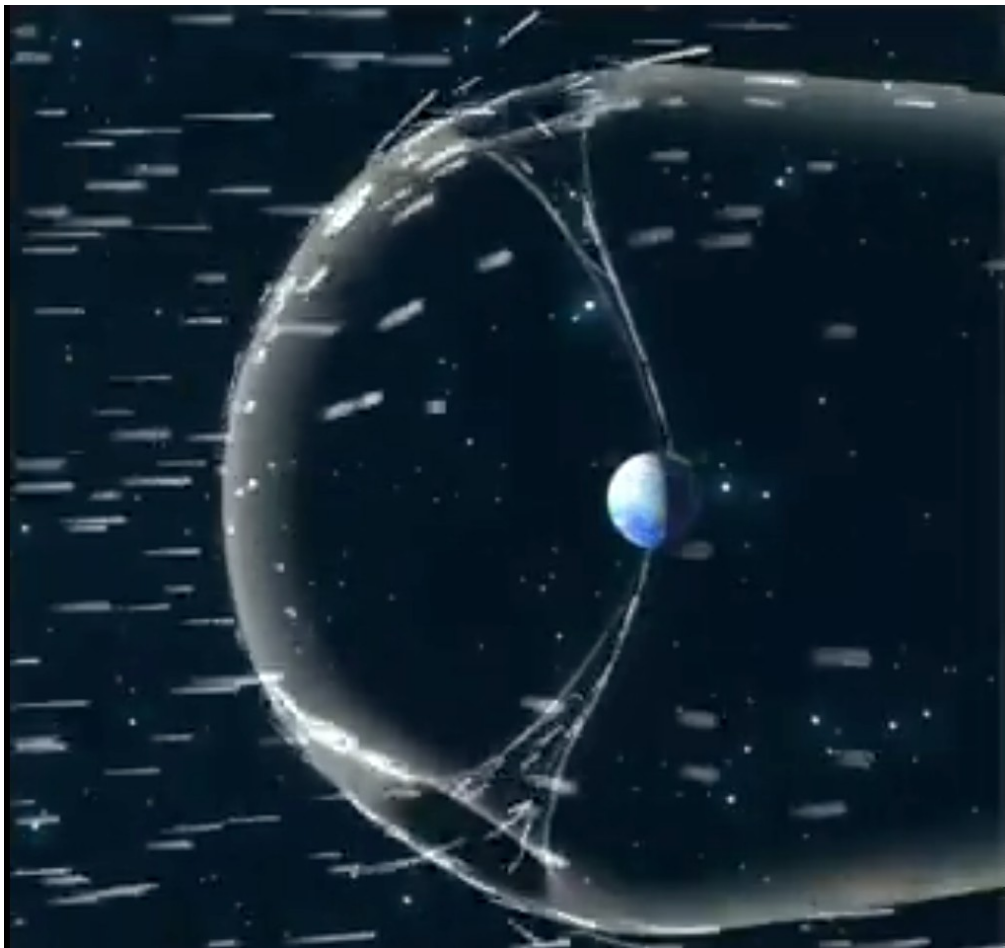




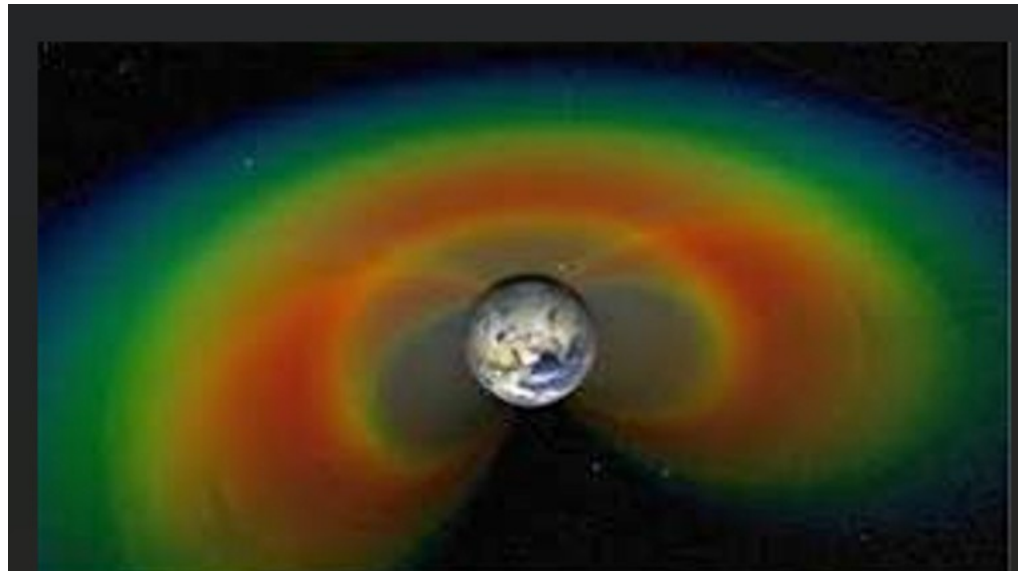
Everything happens
Like the Earth has
A bar magnet inside



Show the Earth magnetic field (Sims) . Earth is like a bar magnet.
<https://www.geologyin.com/2017/07/nickel-is-crucial-for-earths-magnetic.html>

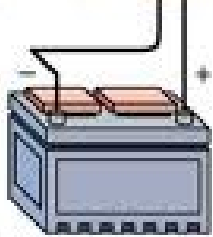
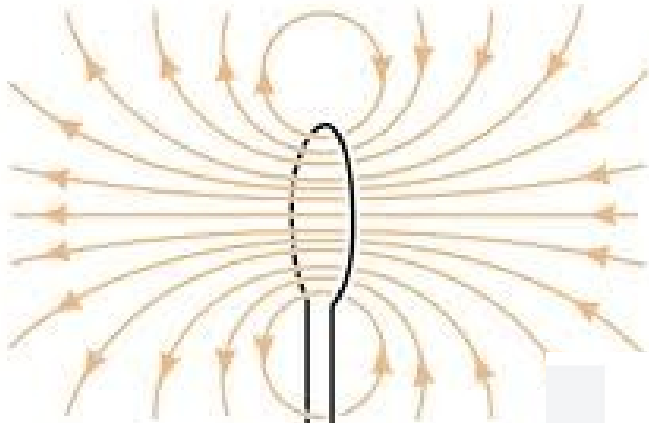


The magnetic field of the Earth
Is like a shield. It protects us
From cosmic rays. Charged
Particles.

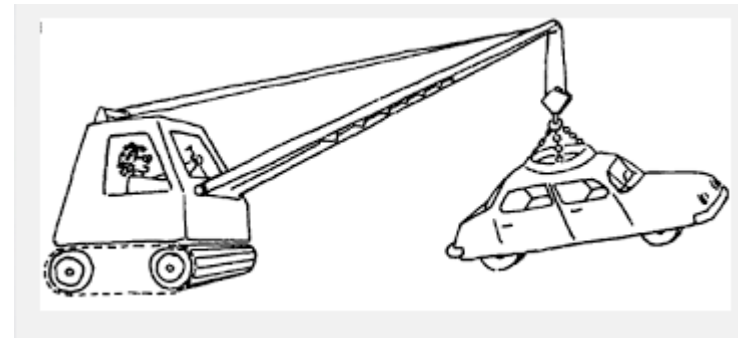
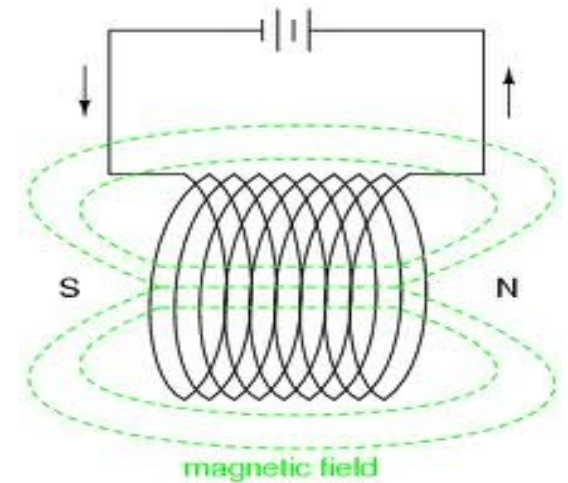


Source of magnetism is electricity.

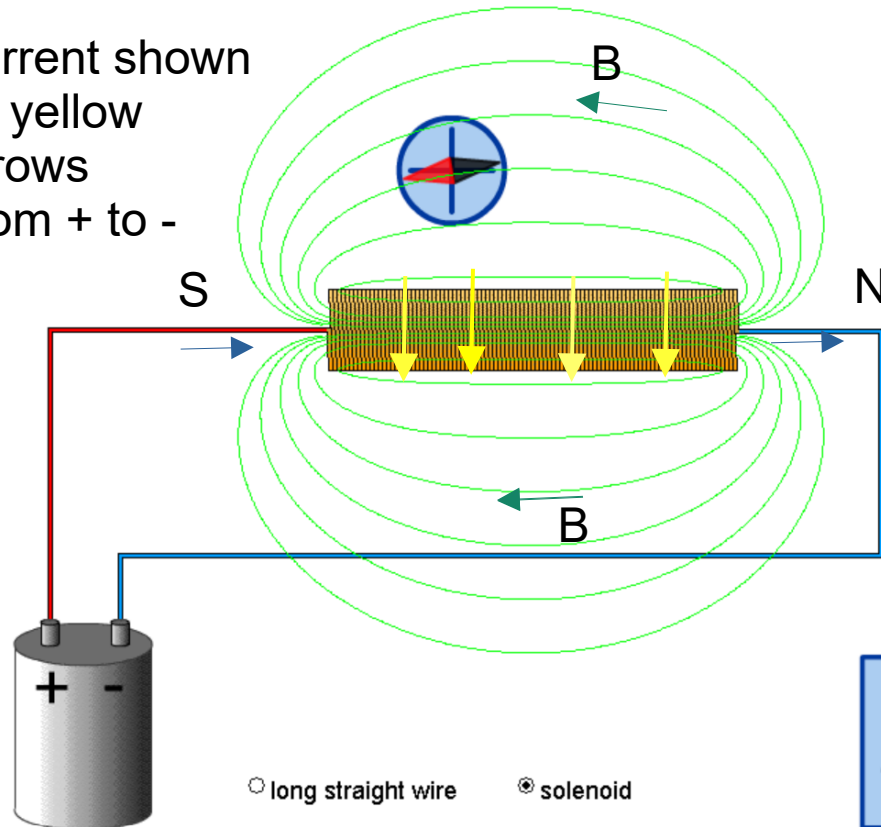
Current in a wire produces a magnetic field in the space around. If the wire is coiled to form loops, the magnetic field is like the ones produced by a bar magnet. The coil becomes an electromagnet with a North and South poles!!



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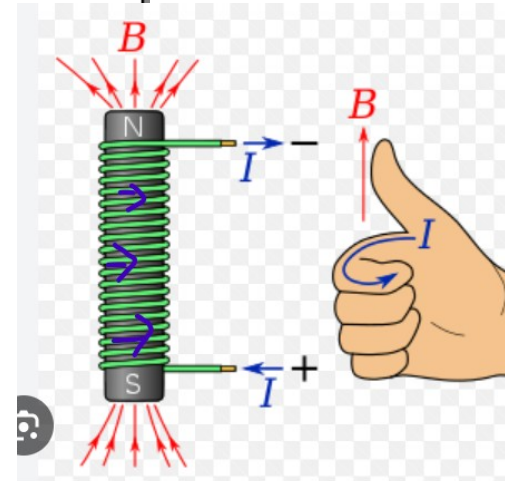
Current shown
By yellow
Arrows
From + to -



○ long straight wire ⊗ solenoid

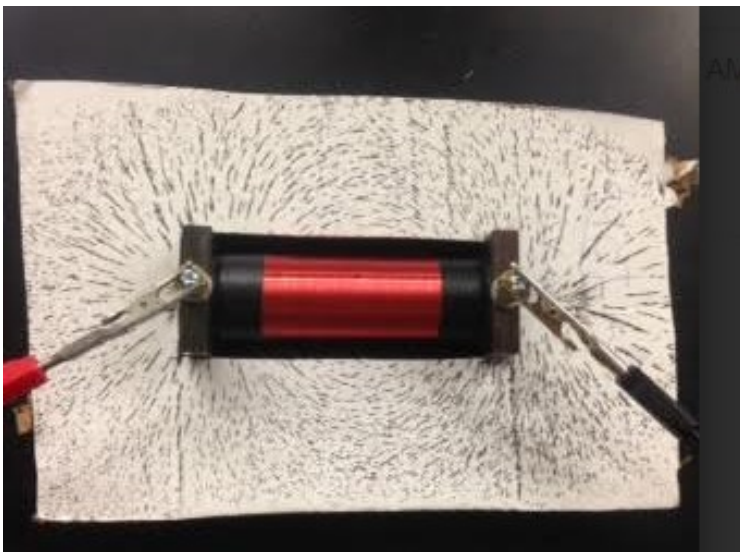
magnetic field:
0.590 T
distance from center:
0.130 m

Use right hand again.



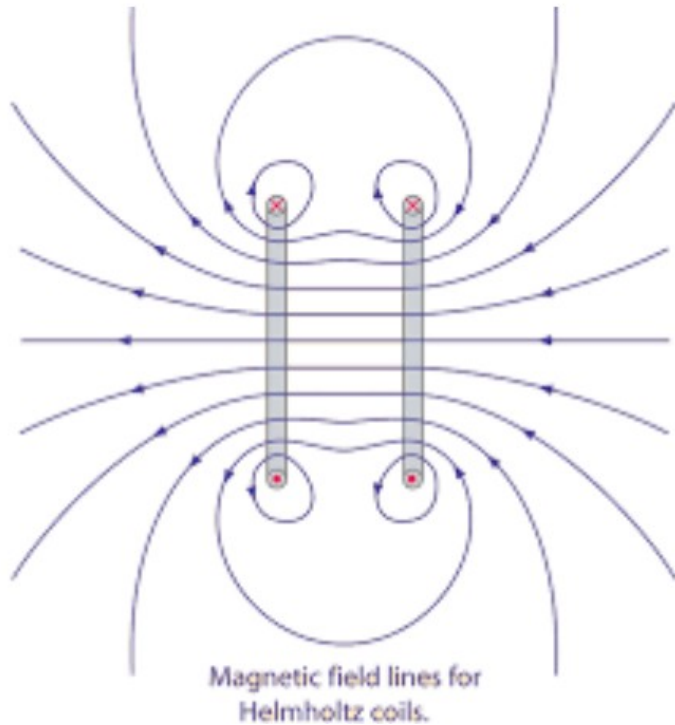
Solenoid;: coil of wire
wrapped tightly
around a cylindrical form

A soft iron core concentrates
The magnetic field inside the
coil



We use iron
Filings to visualize
The field.
(like the seeds for
E)

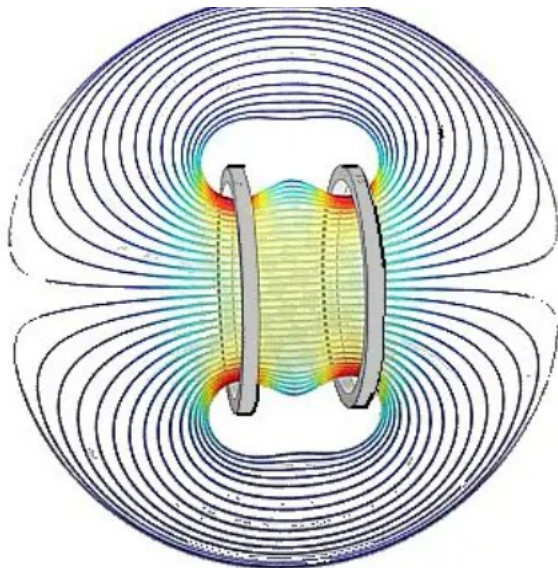
Uniform magnetic field Helmholtz coils



$$B = \mu_0 * N * I / (2R)$$

where:

- B is the magnetic field strength in Tesla (T)
- μ_0 (mu-nought) is the permeability of free space, a constant value of approximately $4\pi \times 10^{-7} \text{ Tm/A}$



Hermotz coils, also called Helmholtz coils, do not produce a perfectly uniform magnetic field, but they get very close under certain conditions.

The uniformity of the magnetic field depends on the distance between the coils. For the most uniform field, the coils

- should be identical circles,
- have the same number of turns, and
- be separated by a distance exactly equal to the radius of the coils https://en.wikipedia.org/wiki/Helmholtz_coil.

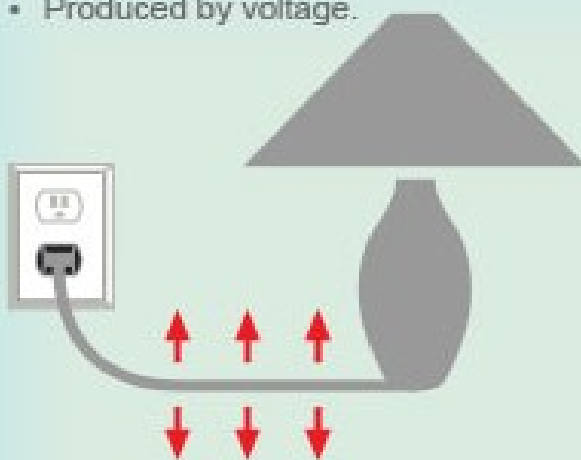
The region of most uniform magnetic field is a cylinder shaped area in the center of the two coils, with a width of about 25% of the coil radius and a length of about half the distance between the coils <https://www.accelinstruments.com/Technical-Info/Technical-Info.html>.

An electrostatic field diverged or converges. Goes from high voltage to low voltage.
A magnetic field. Static or not always curls.

A Comparison of Electric and Magnetic Fields

Electric Fields

- Produced by voltage.

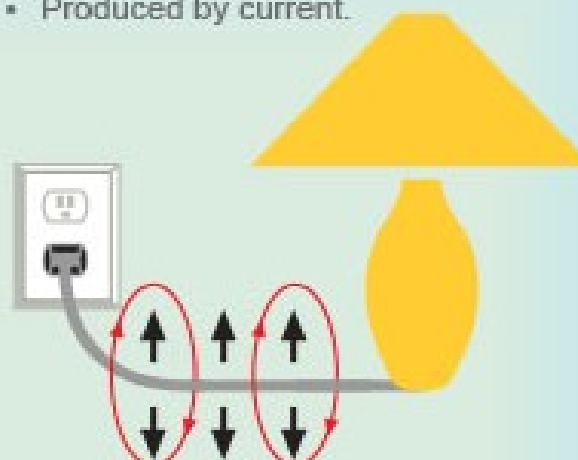


Lamp plugged in but turned off.
Voltage produces an electric field.

- Measured in volts per meter (v/m) or kilovolts per meter (kV/m).
- Easily shielded (weakened) by conducting objects such as trees and buildings.
- Strength decreases rapidly with increasing distance from the source.

Magnetic Fields

- Produced by current.



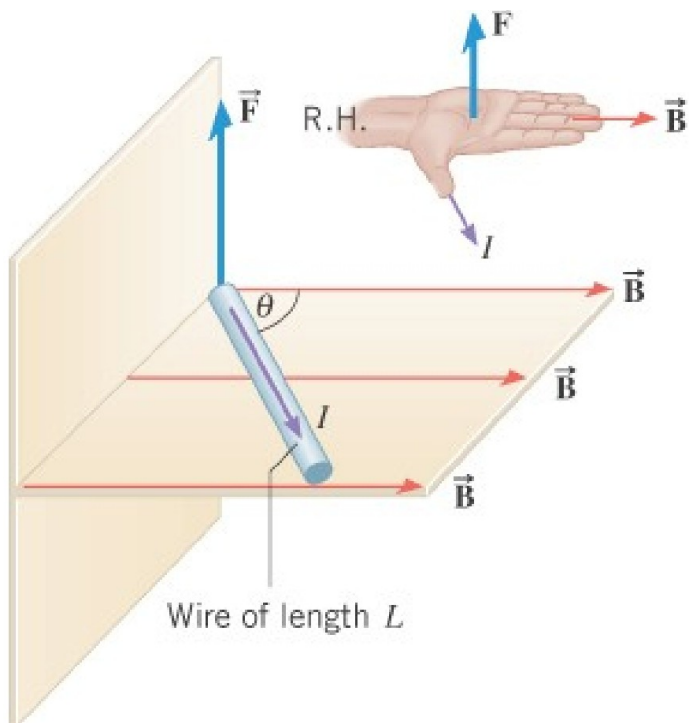
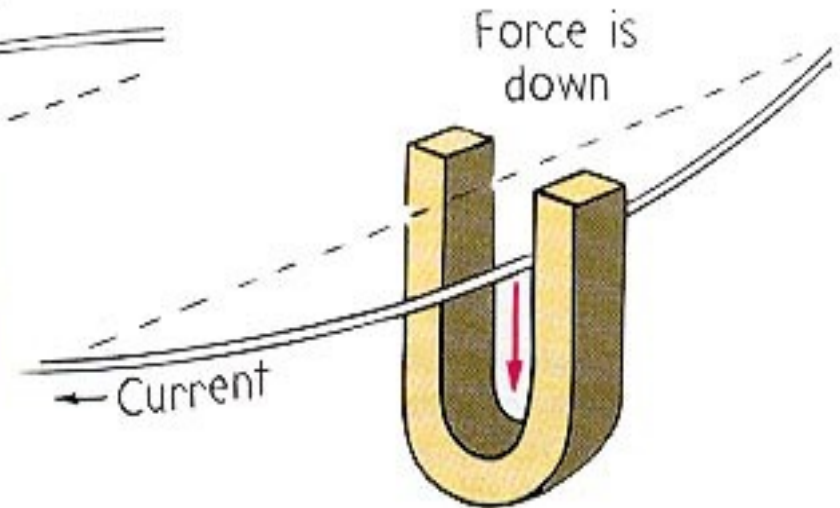
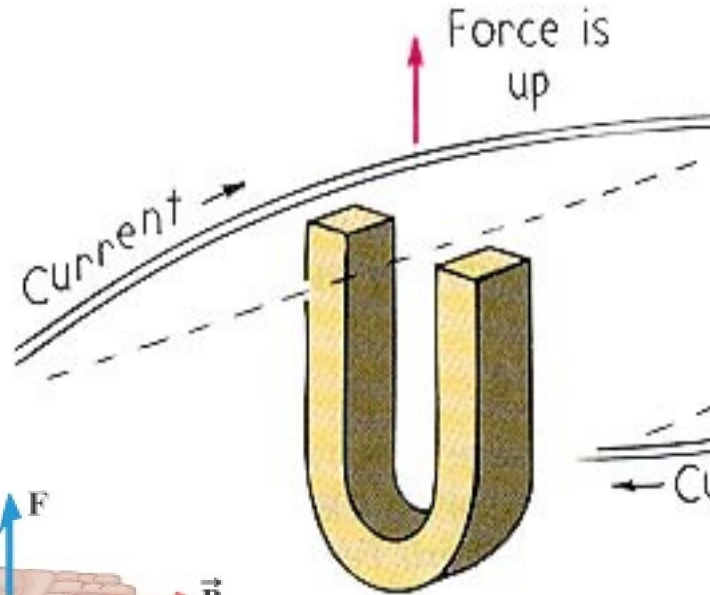
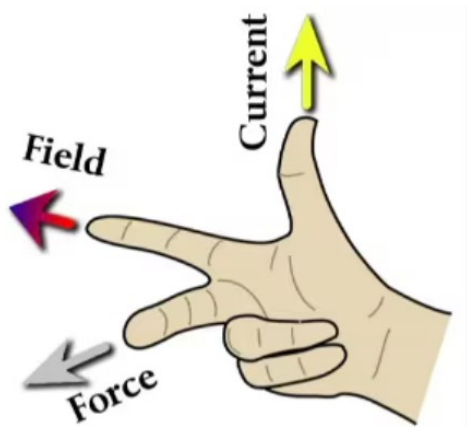
Lamp plugged in but turned on. Current now produces a magnetic field also.

- Measured in gauss (G) or tesla (T)
- Not easily shielded (weakened) by most material.
- Strength decreases rapidly with increasing distance from the source.

A current carrying wire feel a force inside a magnetic field

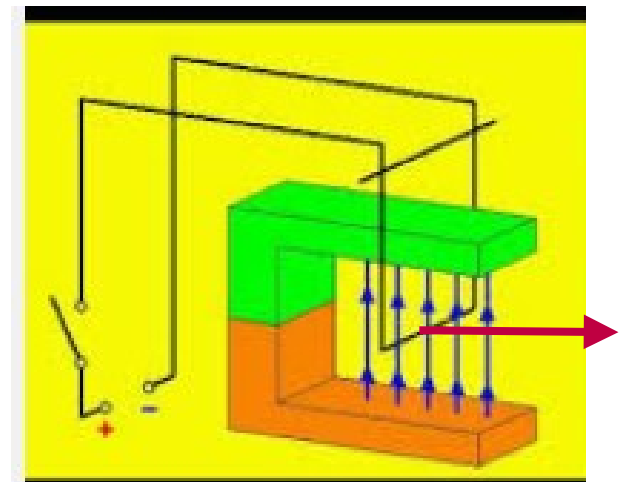
There is no force if I and B parallel or anti parallel

The force produced is perpendicular to both the field and the velocity.

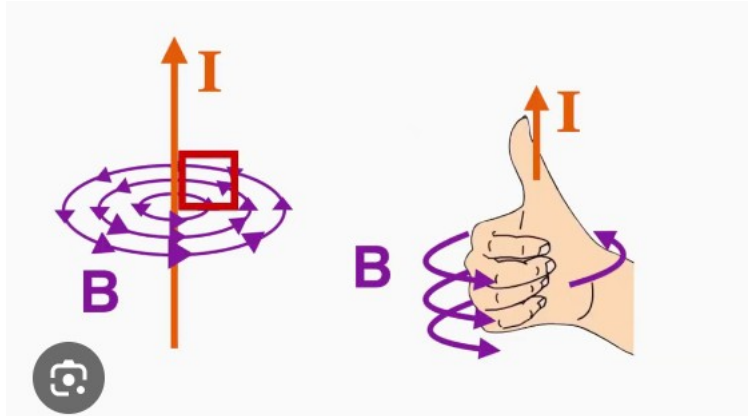


A wire carrying current I , in magnetic field, is acted upon by a force. F is perpendicular to both I and B . if $I \parallel B \rightarrow F=0$

$F = I L B \sin(\theta)$.
Angle between I and B

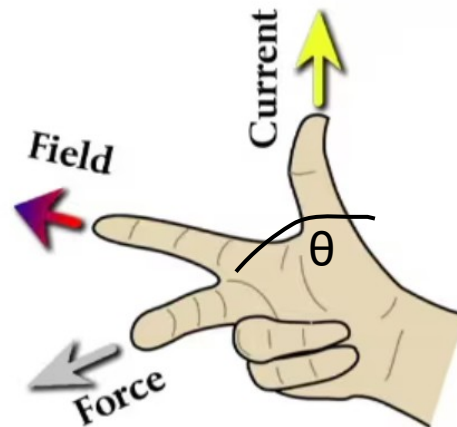
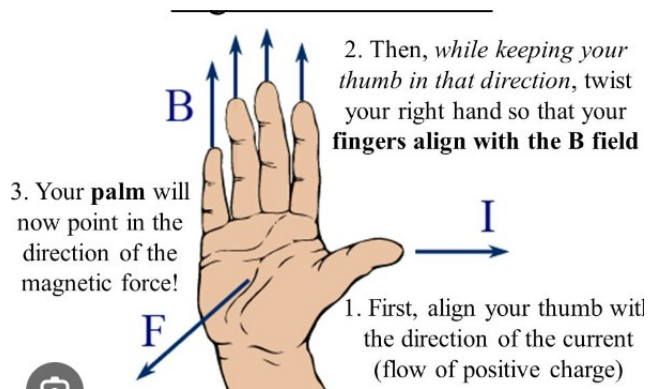
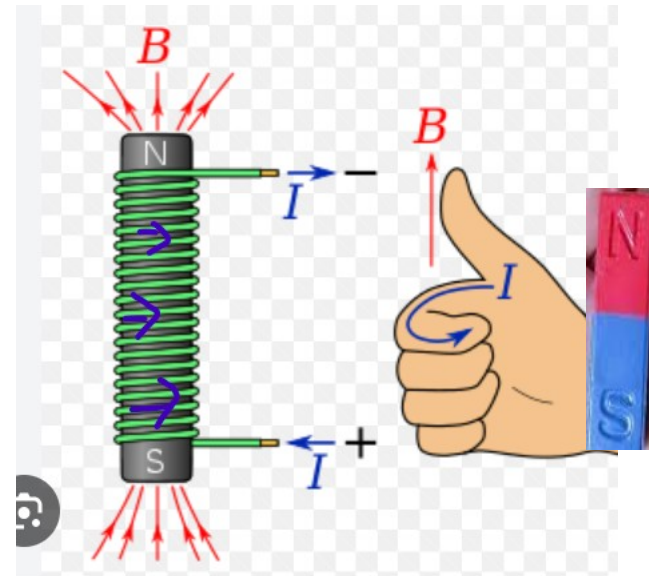


Did you say right hand ?



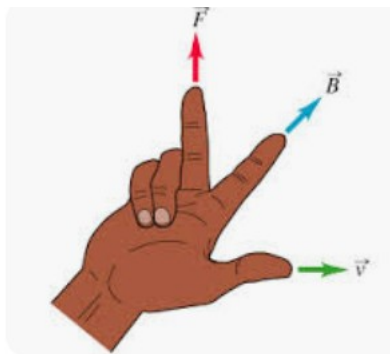
A current carrying wire I
Generate a curling magnetic field.
Use your right hand.

A loop of current behaves like a bar magnet. The thumb shows the north.



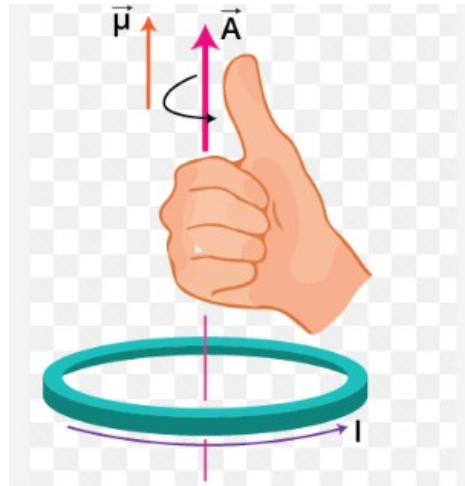
A wire carrying current I , in magnetic field, is acted upon by a force. F is perpendicular to both I and B . if $I \parallel B \rightarrow F=0$

$$F = I L B \sin(\theta).$$



If you have a moving charge q in B . velocity is V . Use same rule: thumb is velocity.
 $F = qVB \sin(\theta)$ θ angle between V and B .
If the charge is negative, force is opposite.

A loop of current or N loops is a magnetic dipole.

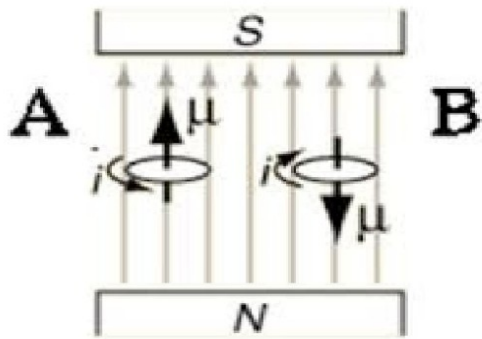
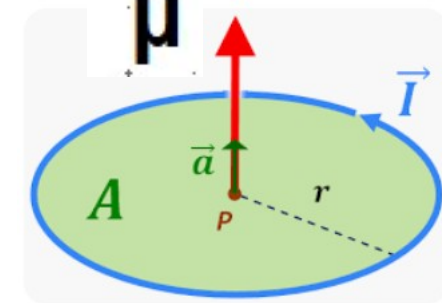


A loop of current behaves like a bar magnet.

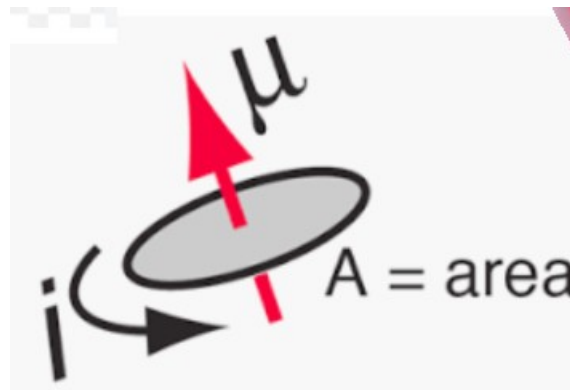
The magnetic dipole moment μ is a vector. It wants to align with B .

$$\mu = N I A$$

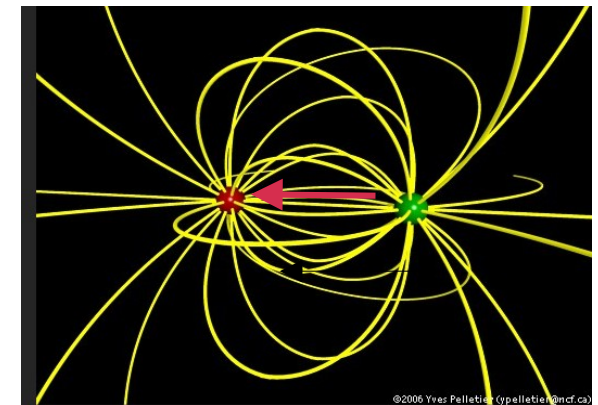
Magnetic dipole μ is perpendicular to area



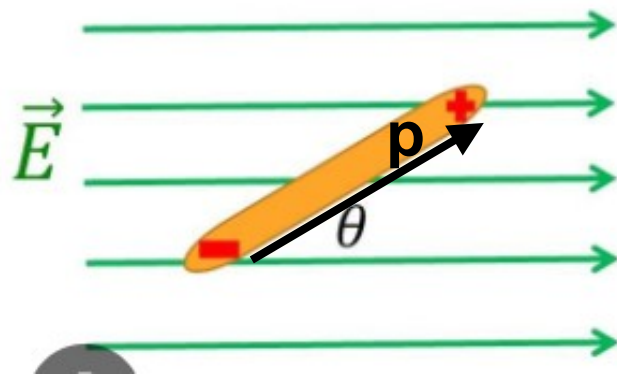
Configuration A has the lowest PE



An electric dipole
The field has a similar configuration
The vector electric dipole $p=qd$



Electric Dipole in a Constant Electric Field



$$\vec{\tau} = \vec{p} \times \vec{E}$$

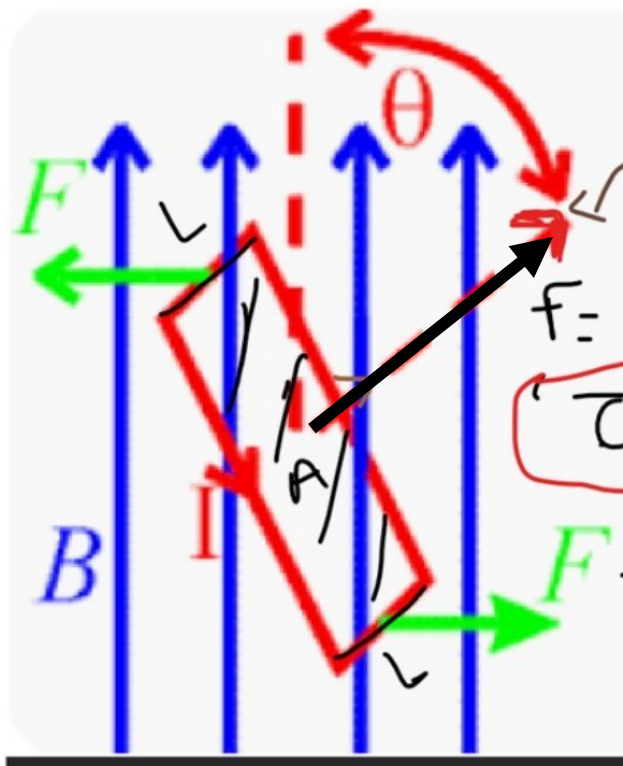
$$U_E = -\vec{p} \cdot \vec{E}$$

$$\mathbf{p} = q \mathbf{d}$$

Electric dipole p is from
- to plus

$$\rightarrow \text{torque} = q d E \sin(E, p)$$

$$\rightarrow U = - q d E \cos(\theta)$$



Vector μ
Wants to align with B

$$F = N I L B \sin(\theta)$$

$$\tau = N I B A \sin(\theta)$$

f. (force) in N
 τ (torque) in N.m
A (area loop) in m^2
N (number loops)

$$\mu = N I A$$

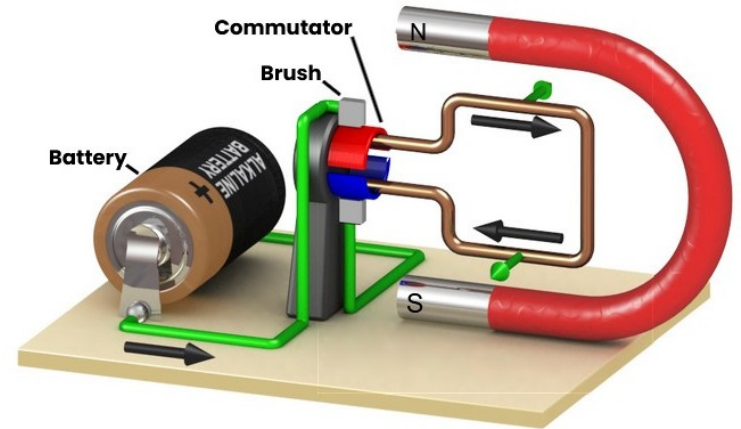
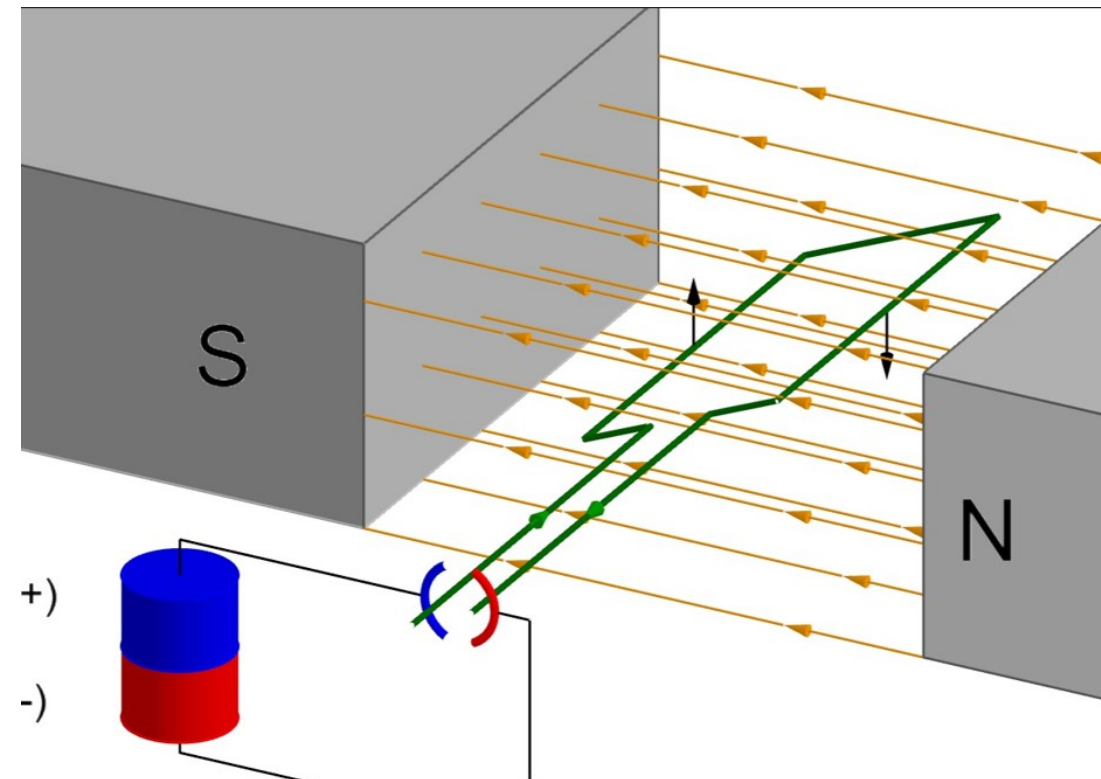
Magnetic dipole μ is
perpendicular to area

$$\tau = \mu \times B$$

$$\tau = \mu B \sin \theta$$

$$U = - N I A B \cos(\theta)$$

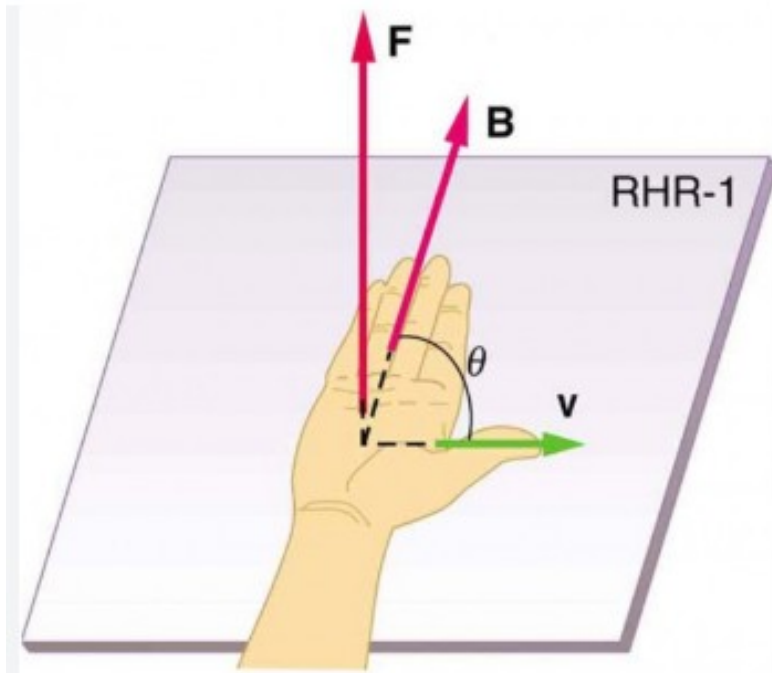
DC electric motor



In DC motors, the commutator reverses the direction of current in the rotor windings every half-turn, creating a steady rotating force (torque)

A moving charge inside a magnetic field feels a force

The force produced is perpendicular to both the field and the velocity.



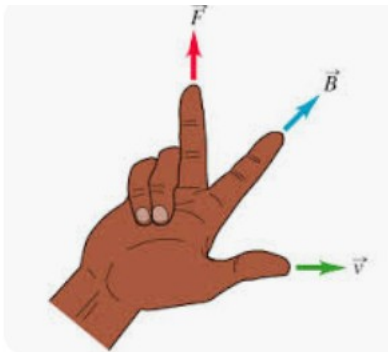
$$F = qvB \sin \theta$$

$\mathbf{F} \perp$ plane of $\mathbf{v}$ and $\mathbf{B}$

If $\mathbf{v} \parallel \mathbf{B}$ (angle = 0 or 180) $F=0$

If $\mathbf{v}=0$ $F=0$

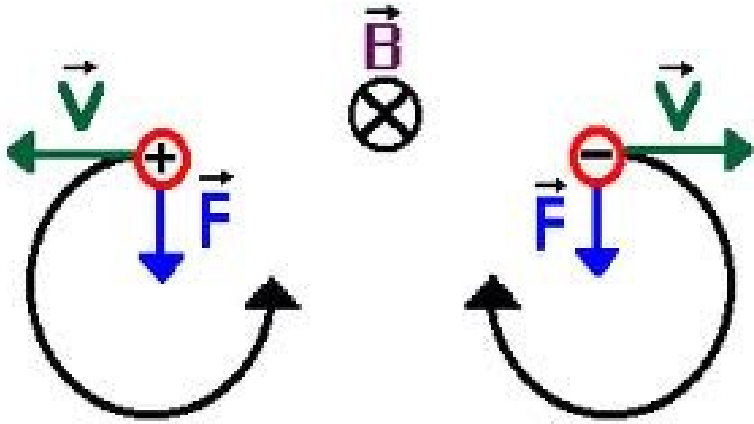
If $q=0$ $F=0$



If you have a moving charge q in $\mathbf{B}$. velocity is $\mathbf{v}$. Use same rule: thumb is velocity.
 $F = qvB \sin(\theta)$ θ angle between $\mathbf{v}$ and $\mathbf{B}$.
If the charge is negative, force is opposite.

A moving charge inside a magnetic field feels a force

The force produced is perpendicular to both the field and the velocity.



The magnetic force deflects the Charge. Its a centripetal force. It keeps the charge on a Circular path.

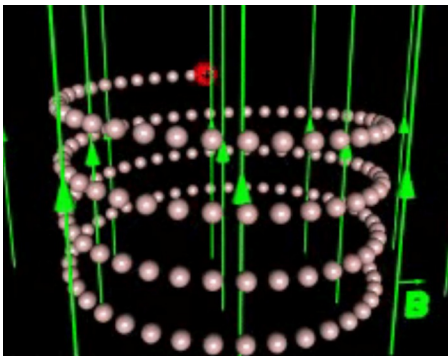
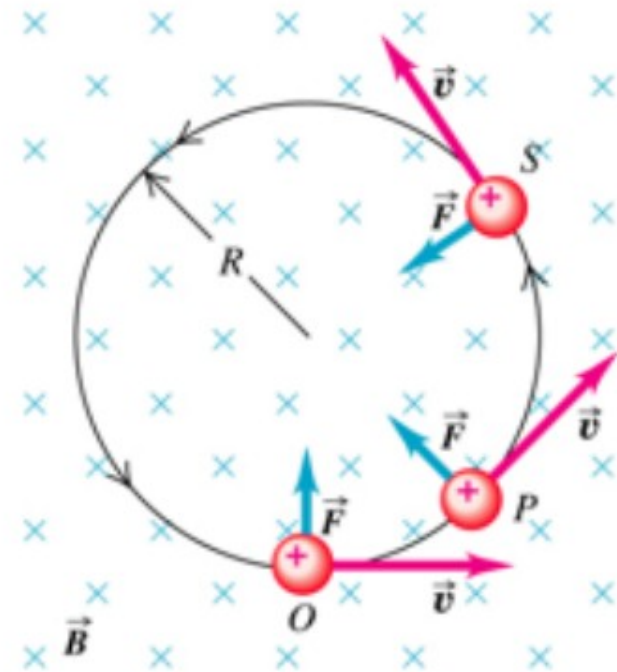
The acceleration is a centripetal acceleration $= mV^2/r$

The radius of the circular path: $r = \frac{mv}{qB}$

The angular speed: $\omega = \frac{v}{r} = \frac{qB}{m}$

The period of the motion:

$$T = \frac{2\pi r}{v} = \frac{2\pi}{\omega} = \frac{2\pi m}{qB}$$



Helical motion. Charged particles circle around the field lines and **radiate**.

cyclotron frequency $\omega_c = qB/m$

Mass spectrometer

1. isotopes are ionized

2- isotopes are accelerated in a E

$$.qU = 0.5 m V^2$$

U is the voltage across the electrodes

3- velocity selector

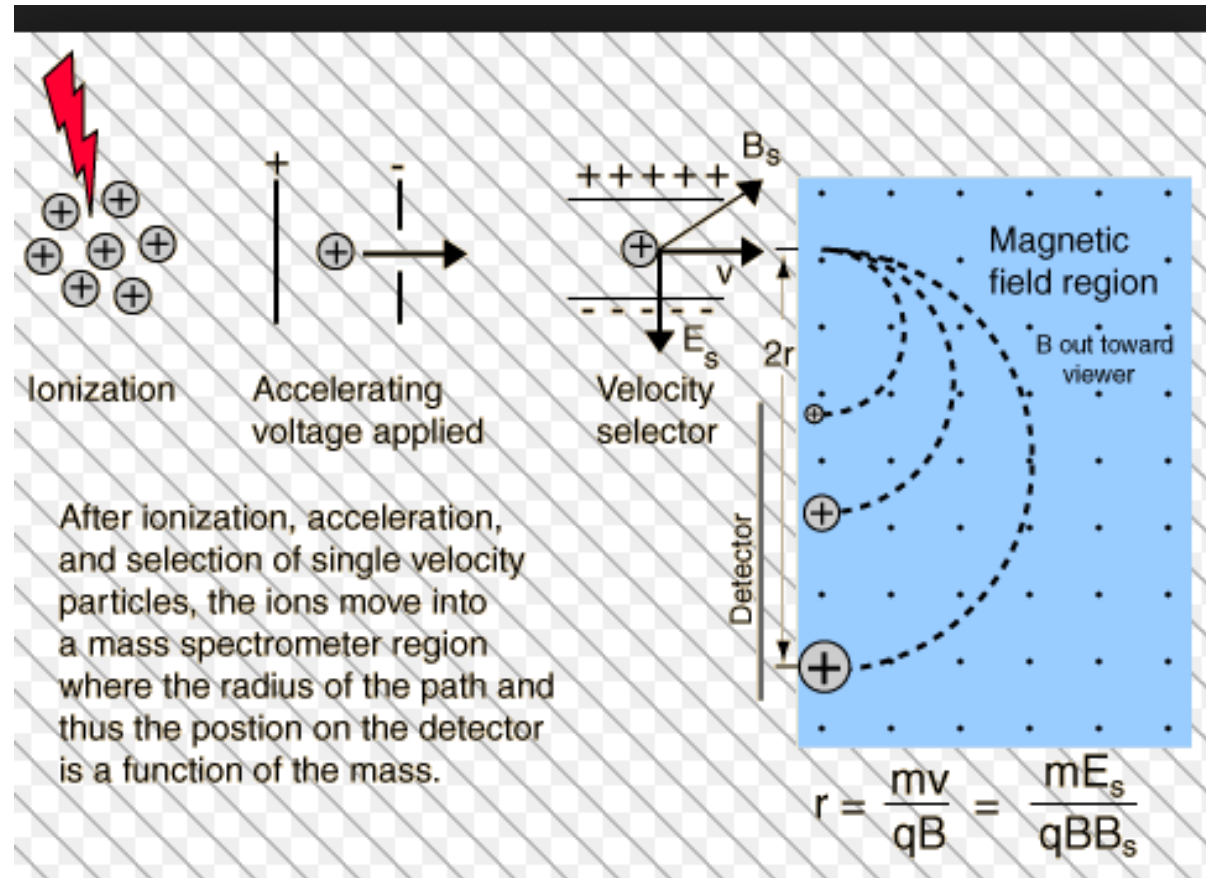
$F_m = qVB$ is up and $F_E = qE$ is down

$$V = E/B$$

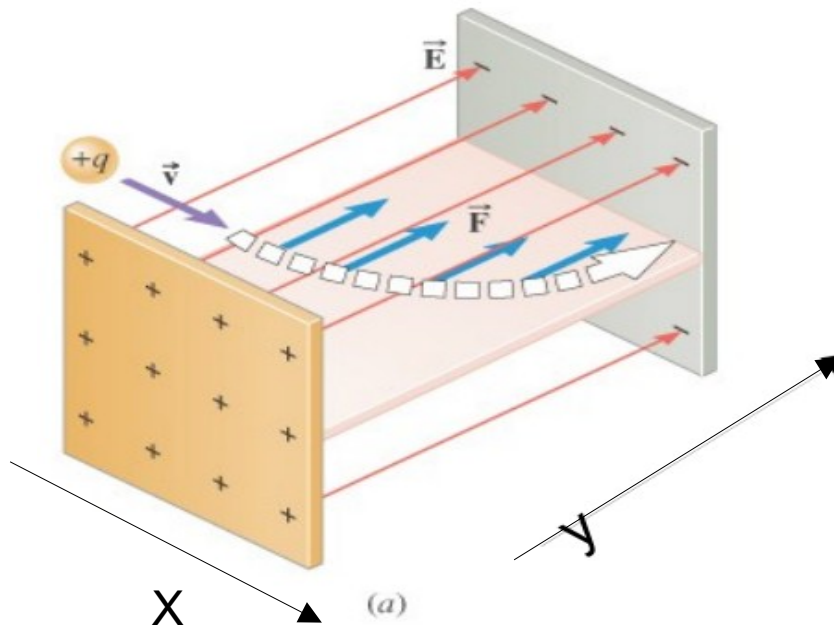
All the isotopes with the same Velocity enter the magnetic field region

4. isotopes have same V, q, B

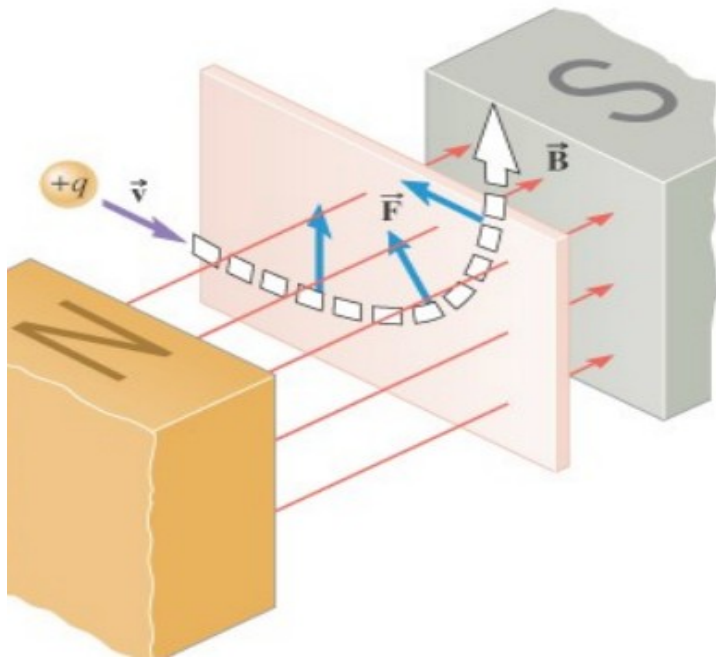
So r is proportional to m only



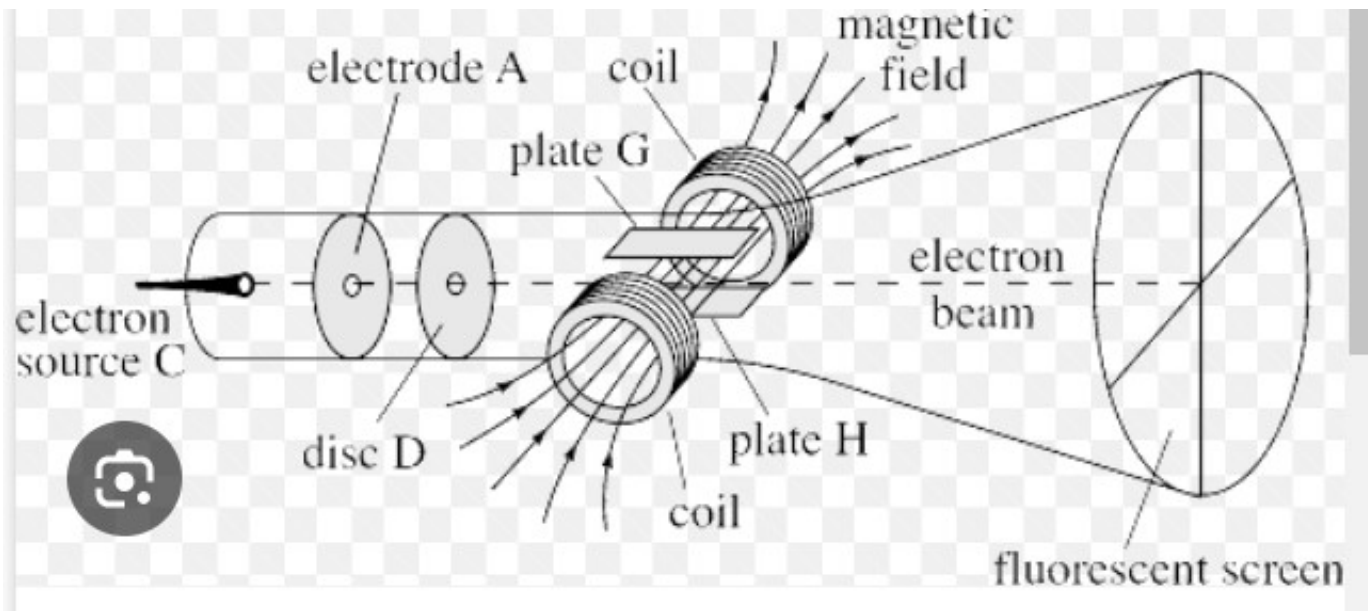
$$m = \left(\frac{qr^2}{2U} \right) B^2$$



An electric field E accelerates a charge q
 $F = qE$ F is along E F is along yaxis
 $a = qE/m$ a is along the yaxis
 V_x stays the same V_y increases
 $V_y = at$

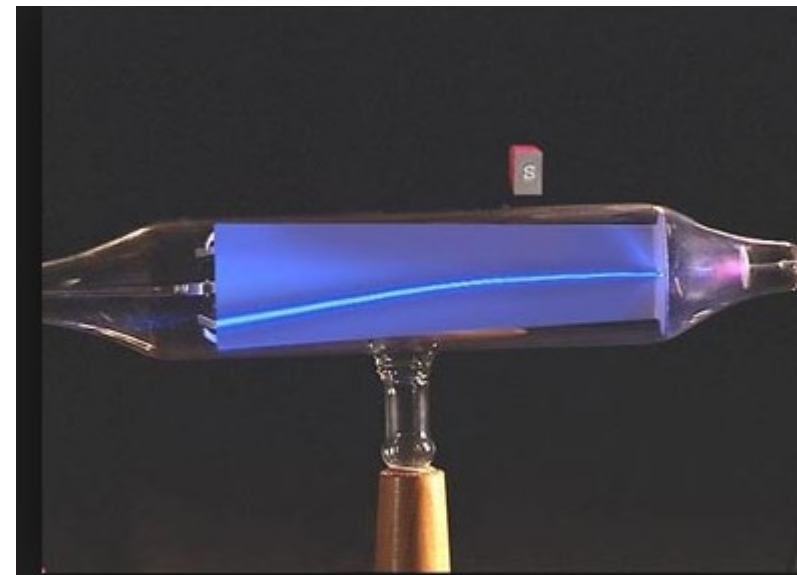


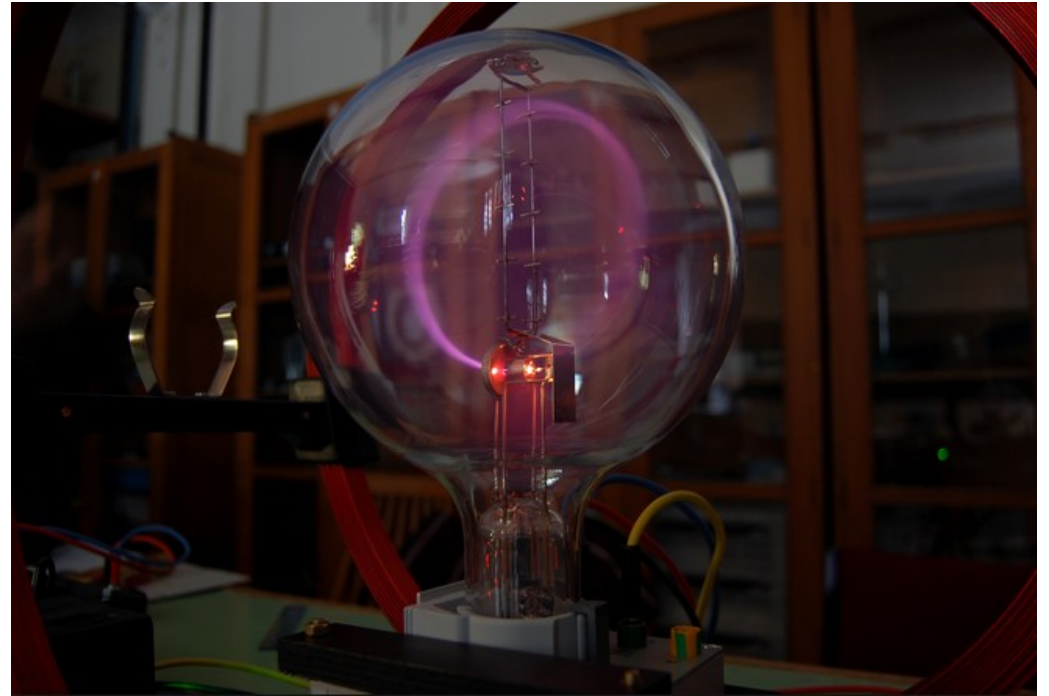
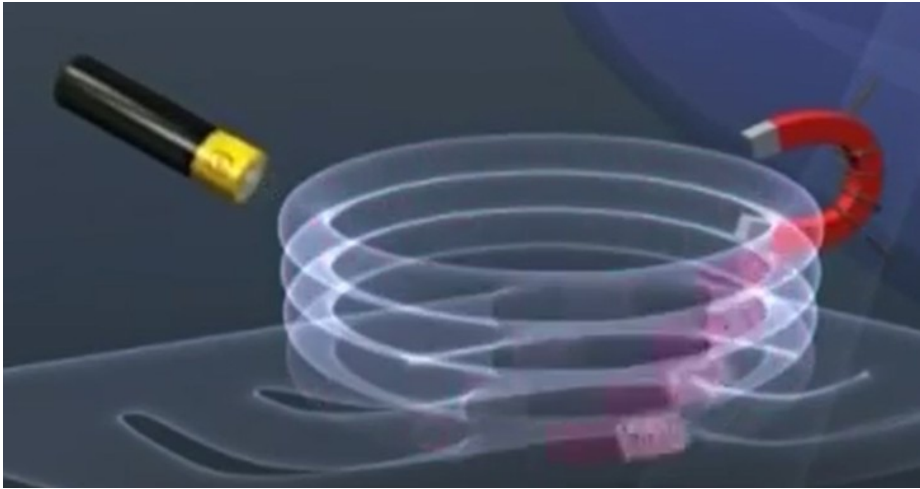
A magnetic field deflects.
 The force $= qvB$ here is a centripetal force
 The acceleration is a centripetal acceleration
 $a = V^2/r = F/m$



The electric field between the cathode and the anode speeds up the Electrons. The magnetic field deflects the electrons.
(idea behind TV screen or oscilloscope)

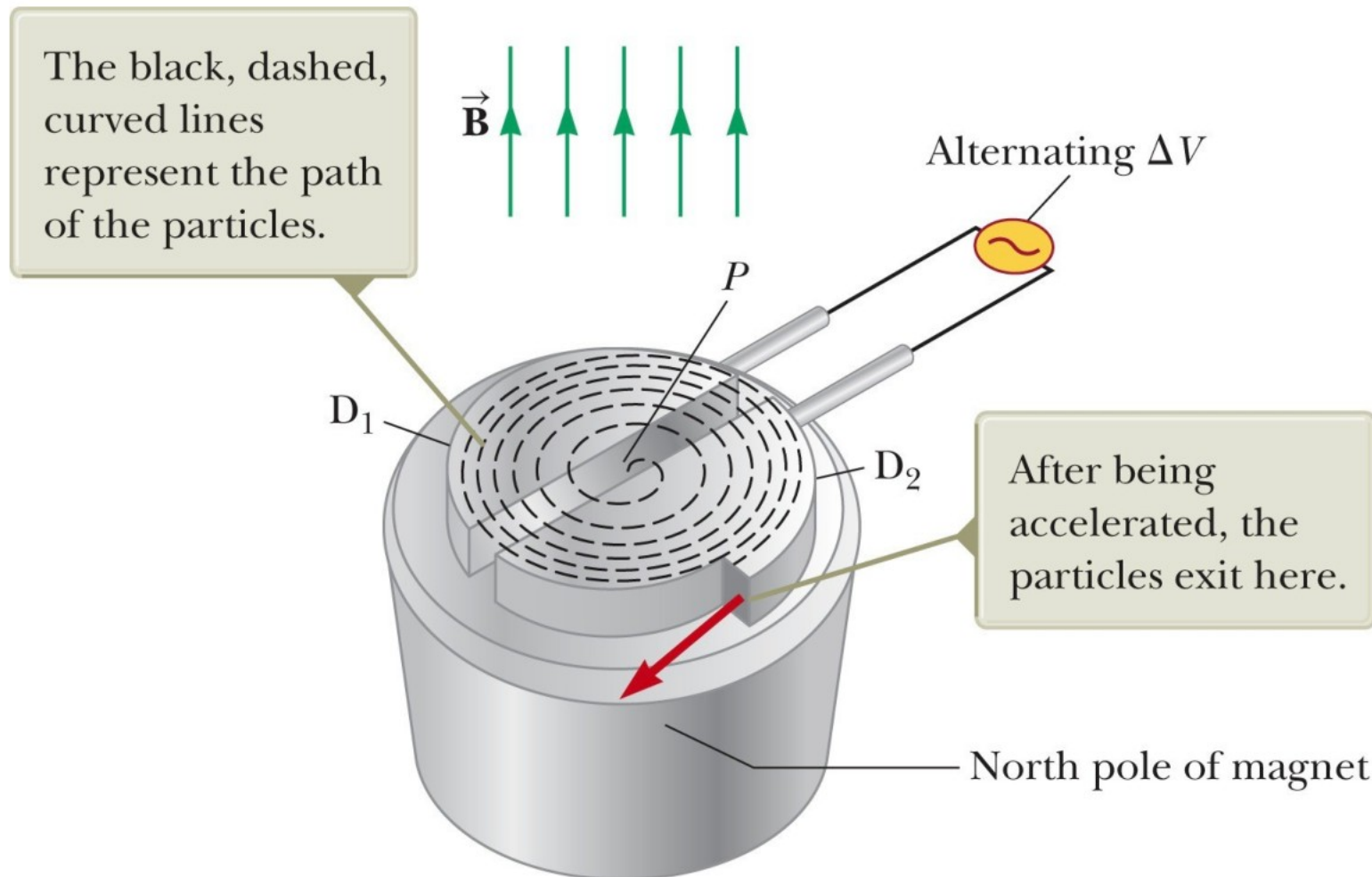
J.J. Thomson built the first mass spectrometer to find the Ration q/m for the electron. (1913).





LHC uses both an electric field and a magnetic field.
E accelerates the charges.
B confine them.

cyclotron



Used for proton
Therapy and to make
radioisotopes

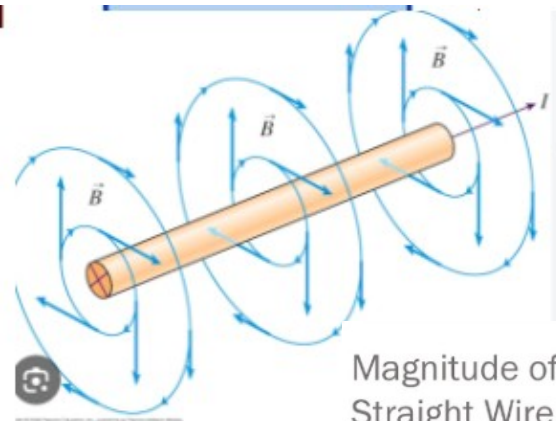
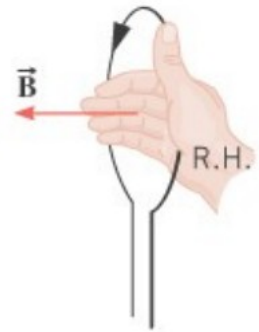
Ernest O. Lawrence built the first cyclotron
(nobel Prize in 1939).

Ampere's law. Source of magnetism

Vacuum permeability, $\mu_0 = 4\pi \times 10^{-7} \text{ (T}\cdot\text{m)/A}$

Magnetic constant, $k' = \mu_0/4\pi = 1 \times 10^{-7} \text{ (T}\cdot\text{m)/A}$

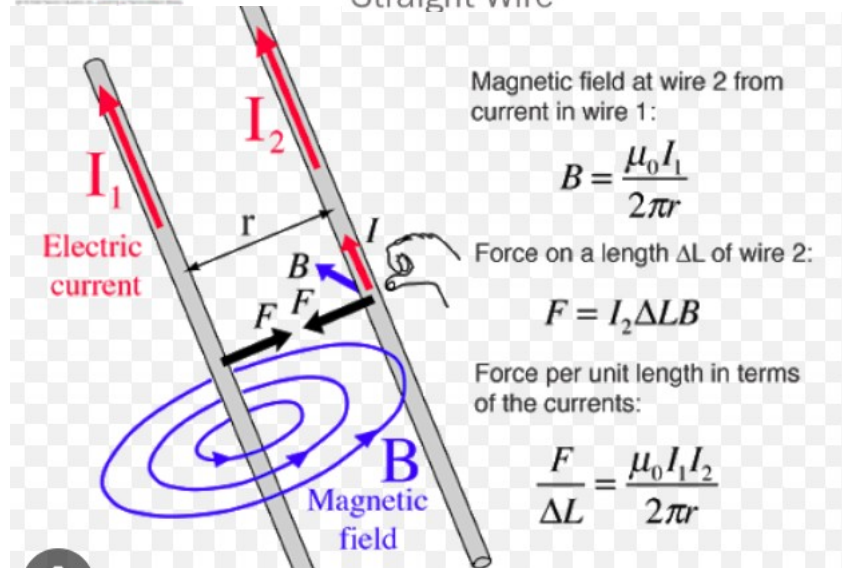
$$B = \frac{\mu_0 I}{2R}$$



$$B = \frac{\mu_0 I}{2\pi r}$$

$B_{\text{wire}} = (2 \times 10^{-7}) I / r$
r is the distance from wire

Magnitude of the Field of a Long Straight Wire



Magnetic field at wire 2 from current in wire 1:

$$B = \frac{\mu_0 I_1}{2\pi r}$$

Force on a length ΔL of wire 2:

$$F = I_2 \Delta L B$$

Force per unit length in terms of the currents:

$$\frac{F}{\Delta L} = \frac{\mu_0 I_1 I_2}{2\pi r}$$

$F/L = (2 \times 10^{-7}) I_1 I_2 / r$
r is the distance from wire
L is the length of the wires

center of circular loop

$B_{\text{loop}} = (2 \times 10^{-7}) \pi I / R$
R is the radius of the loop

- Experiments show that the magnetic flux density inside a solenoid is

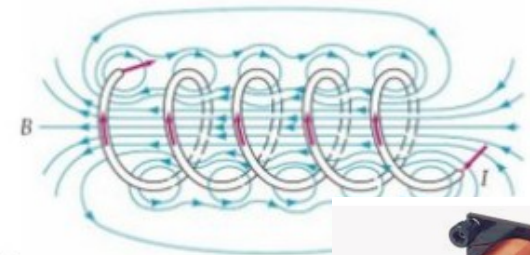
$$B \propto I \quad \text{and} \quad B \propto \frac{N}{\ell}$$

So we have

$$B = \frac{\mu_0 N I}{\ell}$$

$$\text{or } B = \mu_0 n I$$

$$\text{where } n = \frac{N}{\ell}$$



Magnetic flux Φ (Weber) : number of magnetic field lines

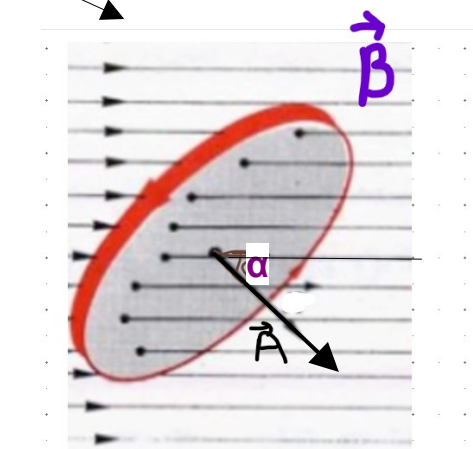
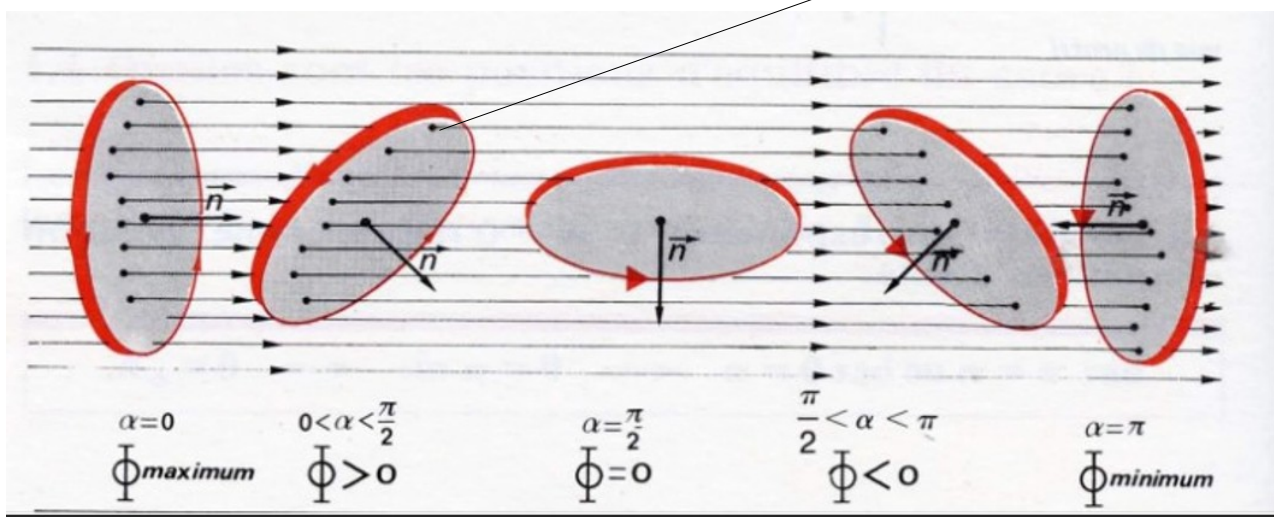
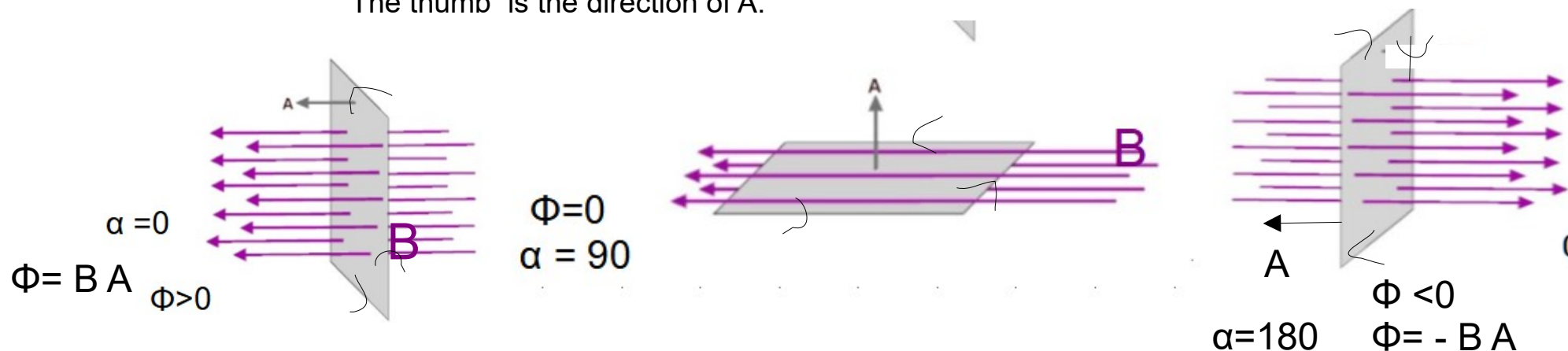
“poking/threading” through “phantom” area A attached to a closed path. Φ is a scalar

$\Phi = B A \cos(\alpha)$ (or $B A_{\perp}$) α angle between A and B

A is a vector perpendicular to area. Magnitude is area

Direction is given by right hand rule. You decide of the direction of the path enclosing the area A .

The thumb is the direction of A .



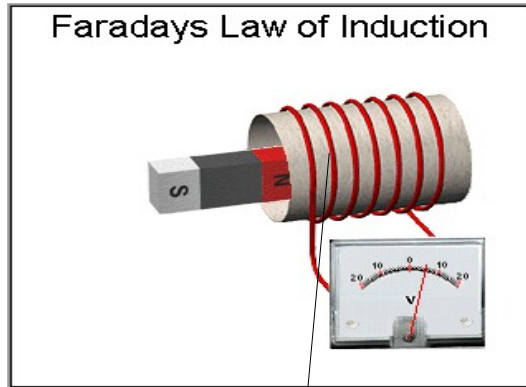
$$\Phi = B A \cos(\alpha)$$

A is a vector. $\vec{n}$ is along A .

Faraday's law of induction

Emf = change in flux through a “phantom” area attached to a conductor (loops)

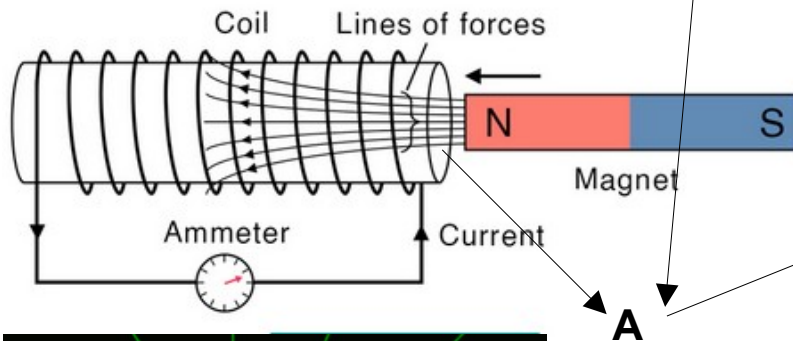
There is an emf if there is a change in the number of lines poking through area A



$$\text{Emf} = (\Delta\Phi/\Delta t) \text{ in volts}$$

$$\text{Emf} = N \Delta(B A \cos(\alpha)) / \Delta t \text{ in volts}$$

Faraday's Law Equation

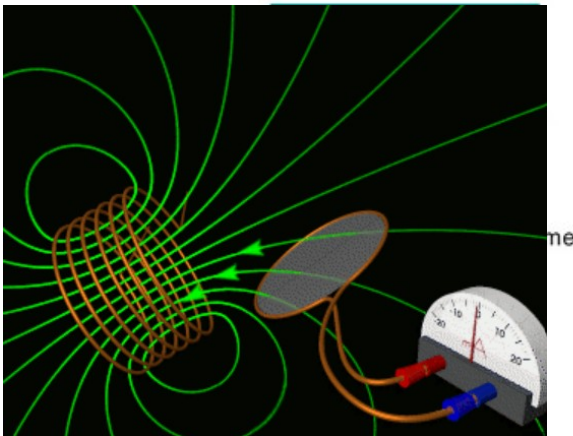


The Flux is the number of field line poking Through area A attached to loops:

$$\Phi = N B A \cos(\alpha)$$

(unit is Weber (Wb) or T m^2)

If B perpendicular to A
 $\Delta\Phi = N (B_2A_2 - B_1A_1)$

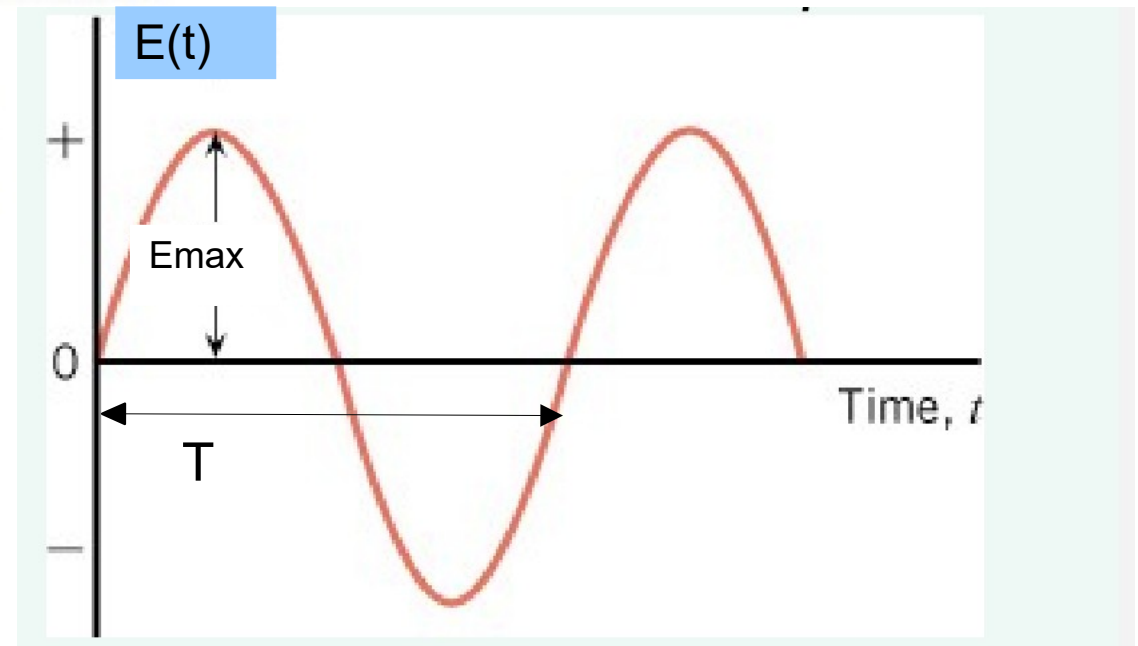
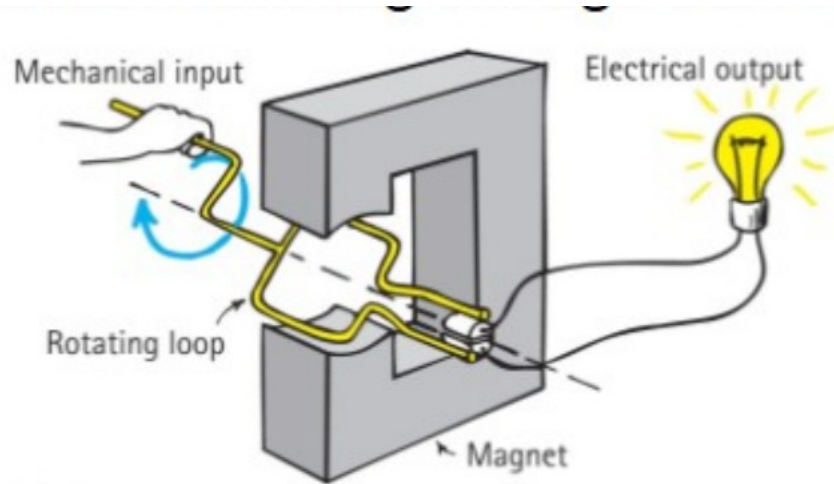


The magnitude of B changes with time. The electromagnet is on and off . There is an induced emf. We can also change the area.

Spin a coil in a magnetic field → get alternating current

An **AC generator** (alternating current generator) is a device that converts mechanical energy (like rotation) into electrical energy in the form of alternating current (AC), where the current periodically reverses direction.

The induced electromotive force (EMF) in an AC generator is alternating (AC)



$$E_{max} = N A B \omega$$

$$\omega = 2\pi/T$$

T is the period of the sine wave.

See next slide

$$E(t) = N A B \omega \sin(\omega t)$$

$$E(t) = E_{max} \sin(\omega t) \text{ with } E_{max} = N A B \omega$$

Consider the voltage $e(t)$ changing with time (t) . Induced by a AC generator.
 The voltage follows a sine function. The equation for a sine function is: $E(t) = E_{\max} \sin(\omega t)$ with

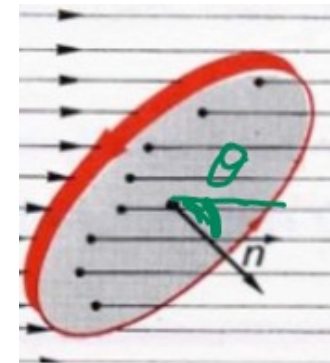
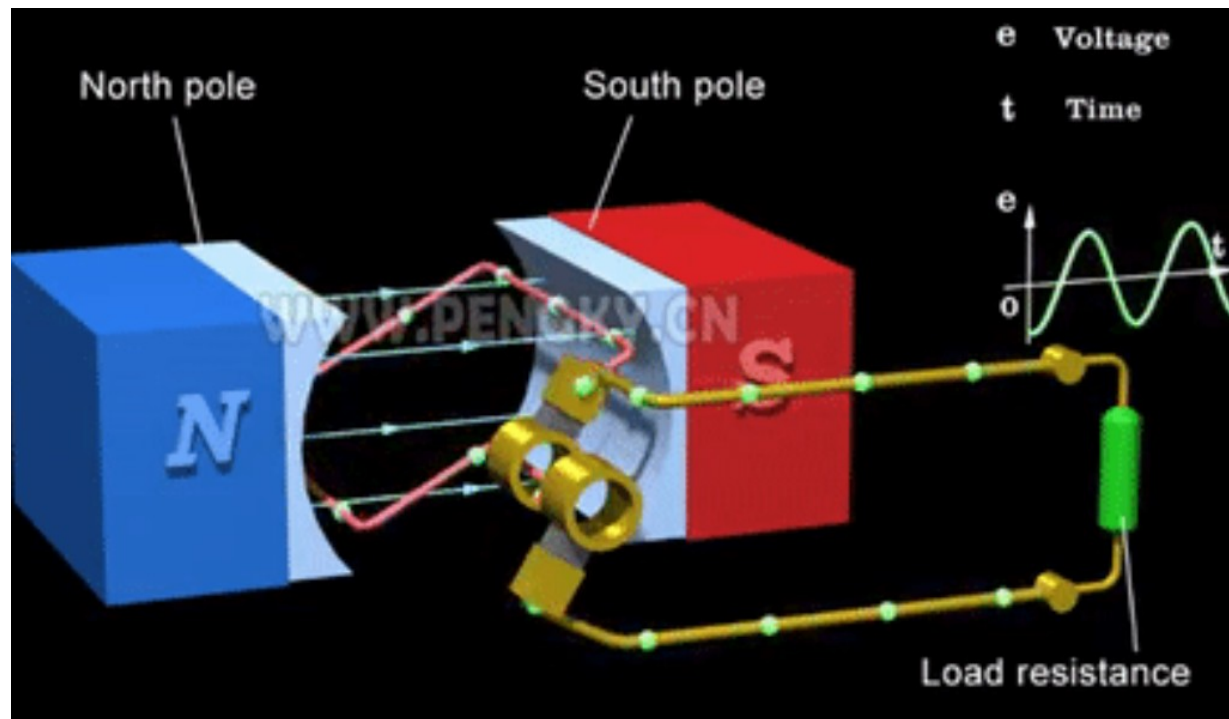
$\omega = 2\pi f = 2\pi/T$. ω is the angular frequency (rad/s)

f is the frequency of the sine function (cycles per second).

T is the period of the sine function (seconds per cycle). $T = 1/f$ or $f = 1/T$.

ω is also how fast the loop is spinning in the magnetic field in rad/s. with $2\pi \text{ rad} = 1 \text{ rotation} = 1 \text{ revolution} = 1 \text{ cycle}$. ω is the rotational speed or the rate of spin of the loop.

$E_{\max} = N A B$ (N number of loops, A area of loop, B magnetic field, ω the rate of spin)



$$\theta = \omega t$$

θ is angle between (n, B)
 n is a unit vector perpendicular to A
 $\theta = \omega t$ (rad)

$$E(t) = N A B \omega \sin(\omega t)$$

In the US $E_{\max} = 170V$

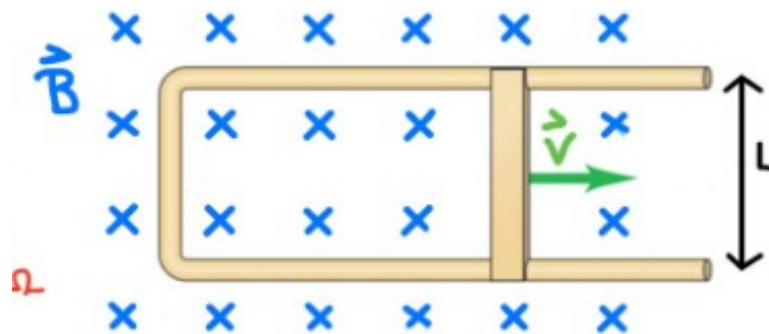
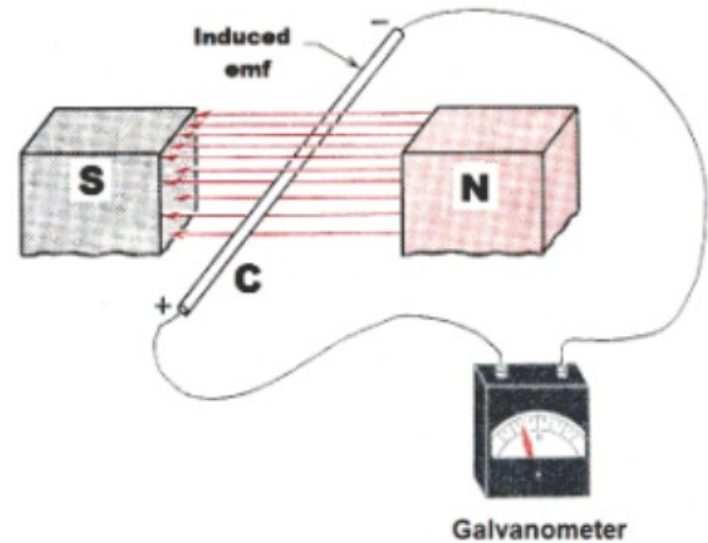
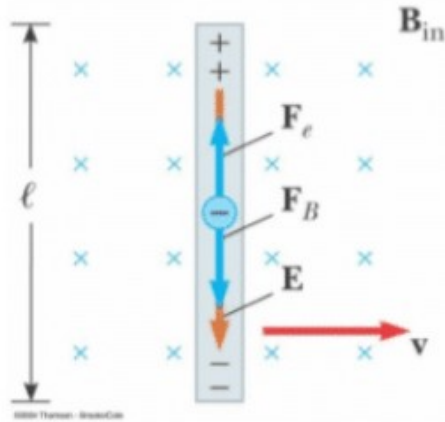
$f = 60\text{Hz}$, $\omega = 2\pi (60) = 120\pi \text{ rad/s}$

$$E(t) = 170 \sin(120\pi t)$$

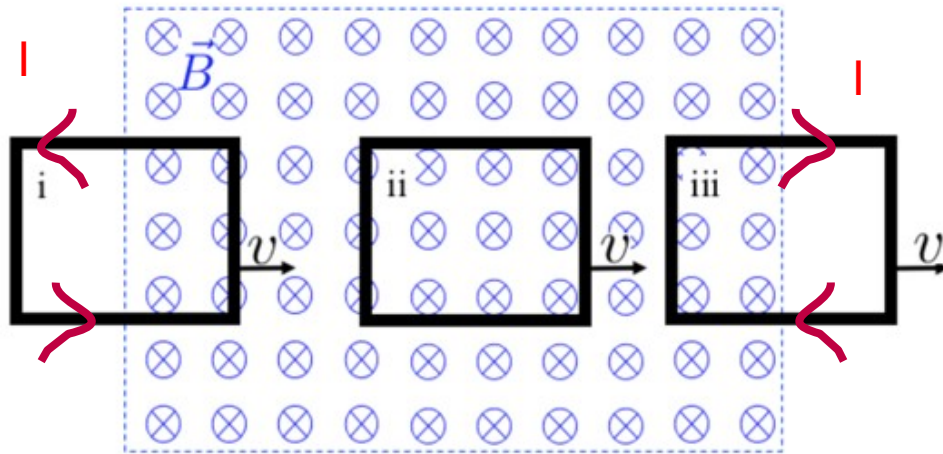
$$\text{Emf} = (N) B L v$$

Motional emf

- A **motional emf** is one induced in a conductor moving through a constant magnetic field
- The electrons in the conductor experience a force, $\mathbf{F}_B = q\mathbf{v} \times \mathbf{B}$ that is directed along ℓ



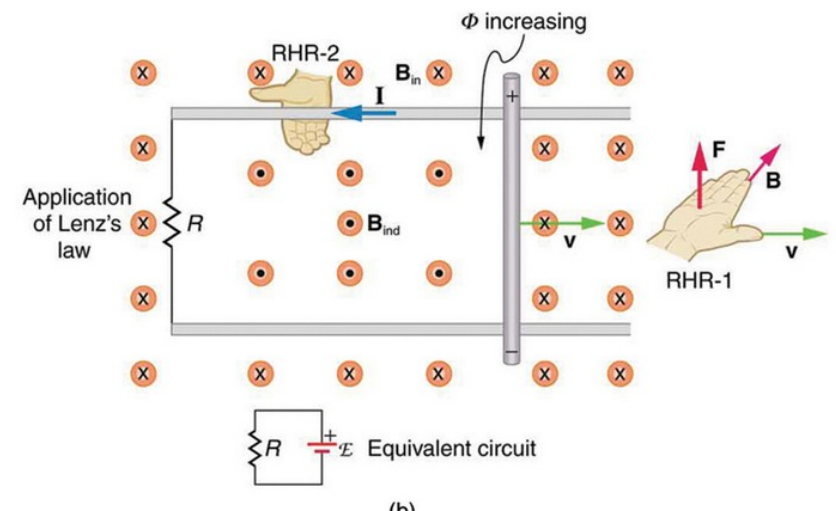
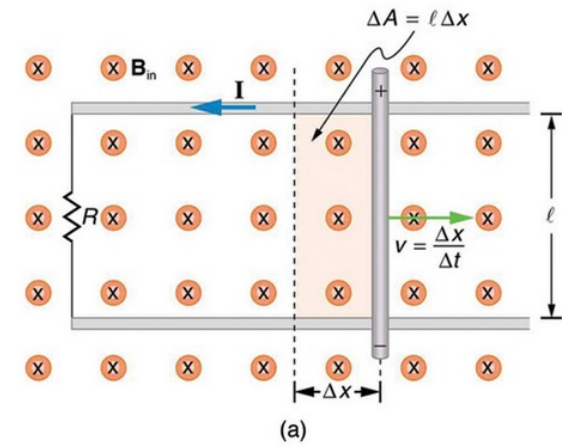
$$\mathcal{E}_{MF} = vLB$$



There is an emf only when the number of lines poking through area A is changing.

So only for .i and iii. $\text{Emf} = v L B$

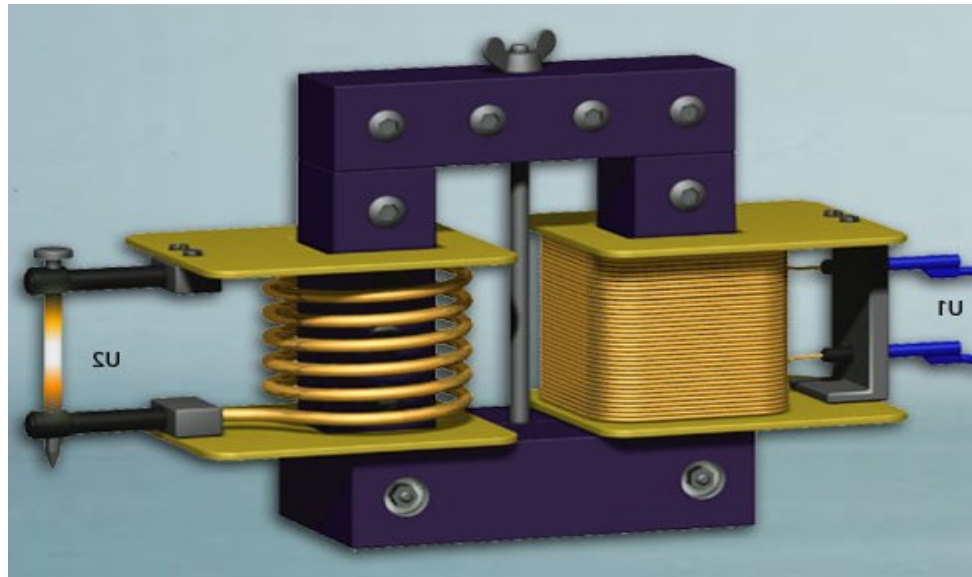
. the current induced $\rightarrow$ use lenz law



transformers

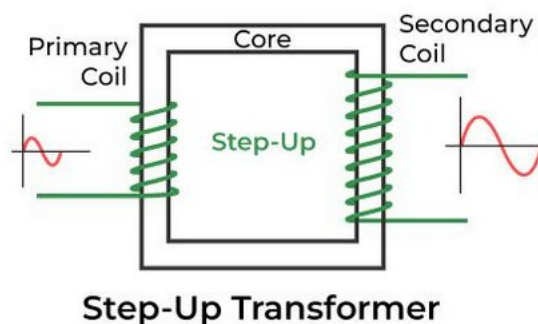
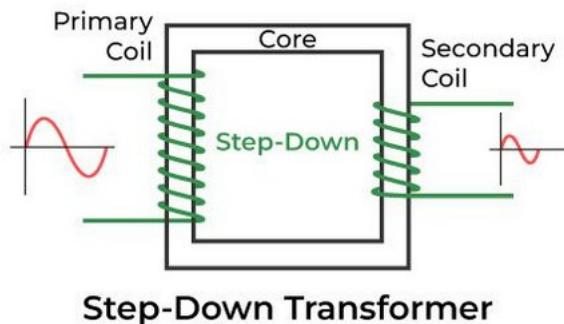
<https://www.youtube.com/watch?v=j6dx6R8YWjk>
demo

$N_1 = 10 \text{ loops}$ $N_2 = 40 \text{ loops}$
 $U_1 = 12 \text{ V}$ $U_2 = 48 \text{ V}$
 $I_1 = 1 \text{ A}$ $i_2 = 0.25 \text{ A}$
 Power = 12W



Transformer : alternative current (1A) in 1 coil (voltage 12V) creates a changing Magnetic field that, guided by the iron square, induces an alternative current in the second coil (more loops). The alternative current is less and the voltage stepped up such as the energy is conserved : $u_1 I_1 = u_2 I_2$

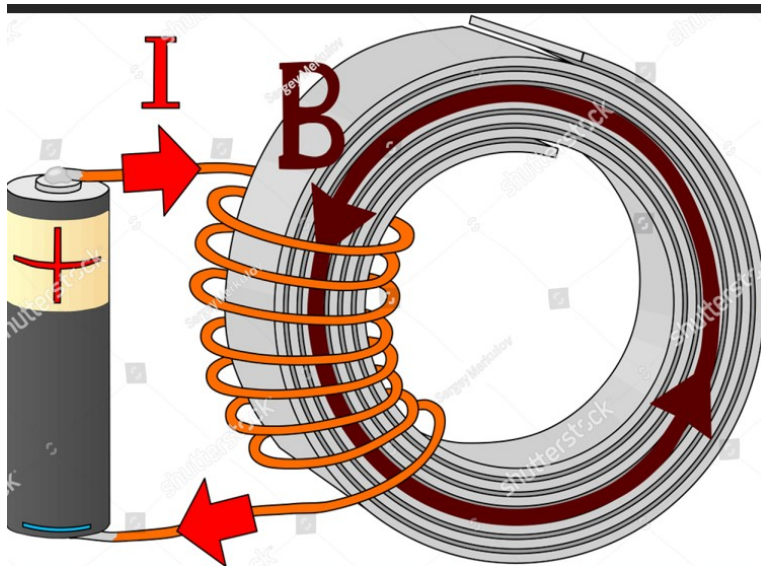
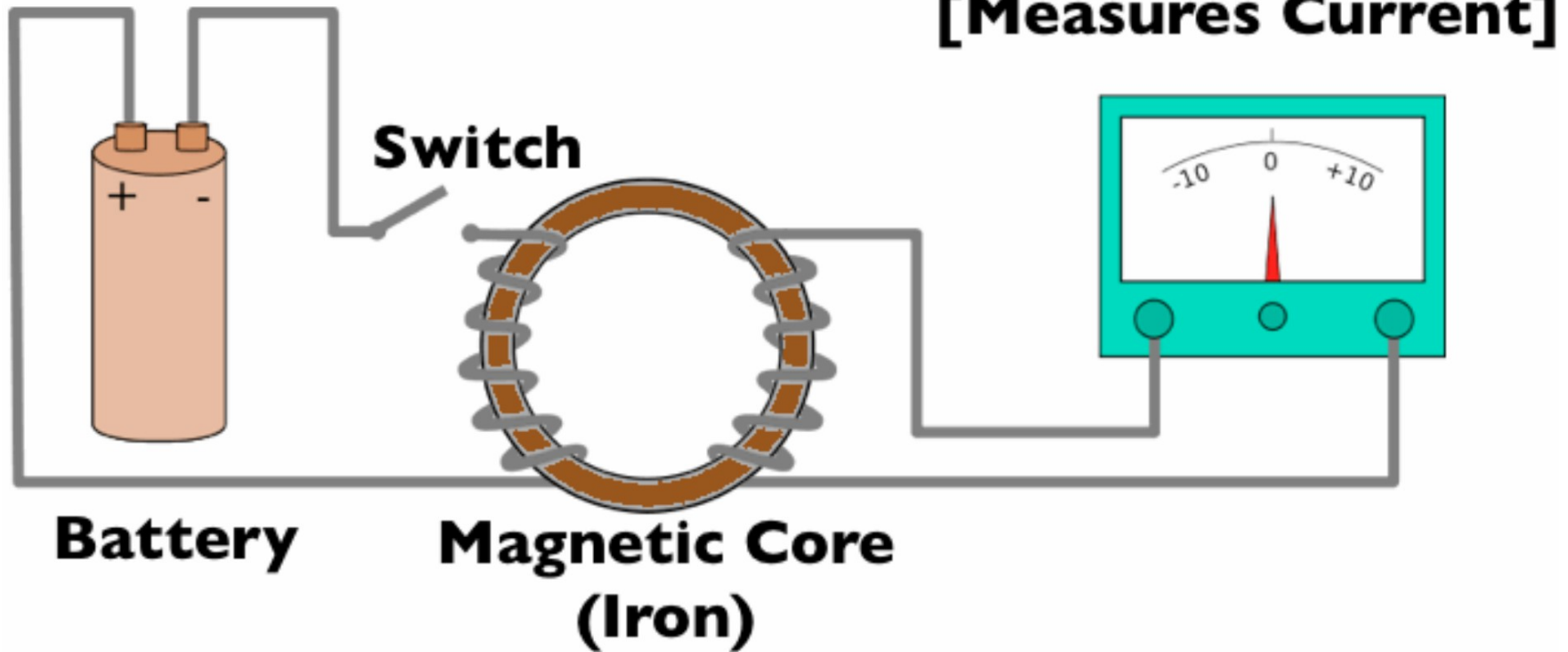
types of transformer



$$\frac{V_P}{V_S} = \frac{I_S}{I_P} = \frac{N_P}{N_S}$$

Experiment done by Faraday. toroid.

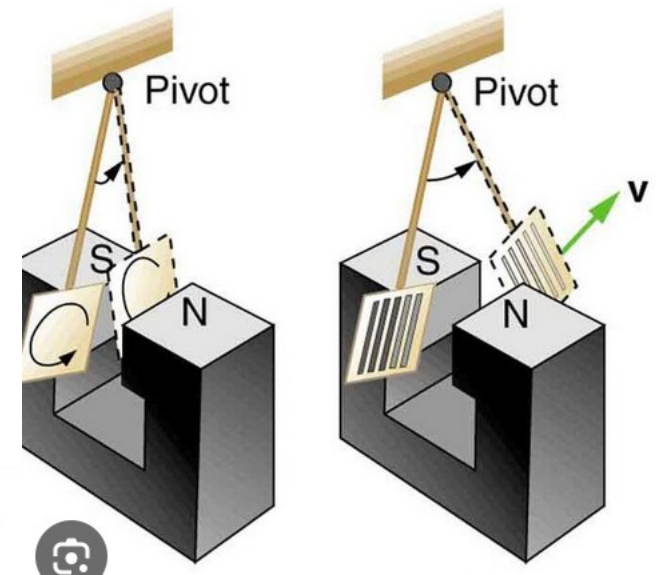
www.maxwells-equations.com



Lenz law: induced current

The induced current opposes the cause that gave rise to it

You induce a current in the loop. The induced current creates a magnetic field that oppose the primary one.



An induced Current always flows in a direction such that it opposes the change which produced it.

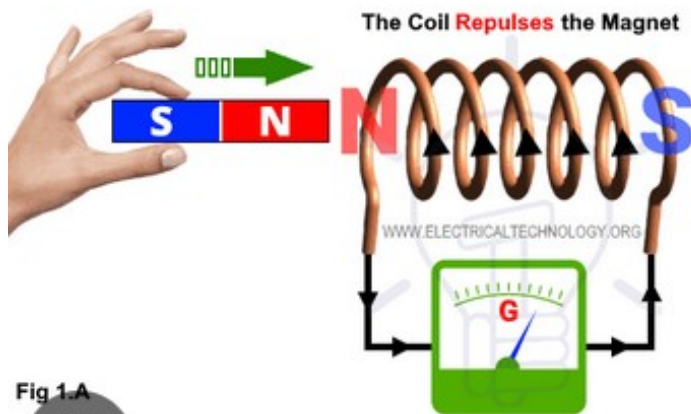


Fig 1.A

When the "N" Pole of the magnet is moved towards the coil, end of the coil becomes "N" Pole.

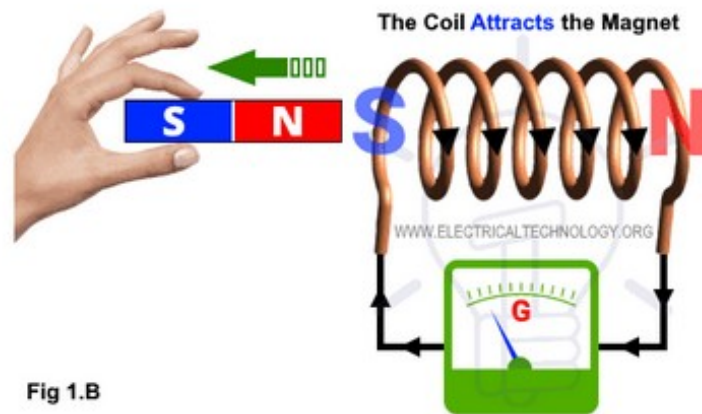


Fig 1.B

When the "N" Poles of the magnet is moved away from the coil, end of the coil becomes "S" Pole.

<https://www.youtube.com/watch?v=sPLawCXvKmg>

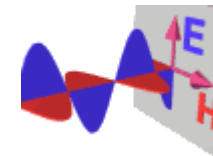
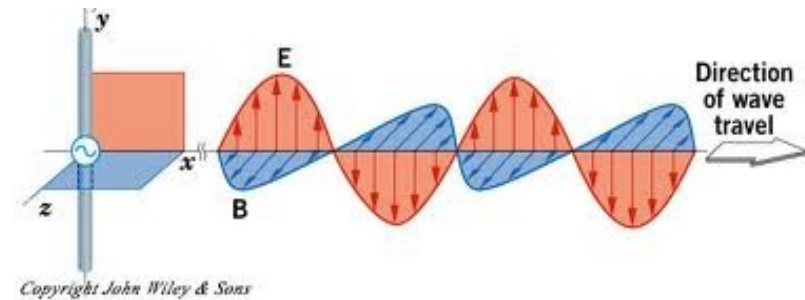
<https://www.youtube.com/watch?v=38XPT9sWlso>

BIG IDEAS IN EM

1) electromagnetic wave

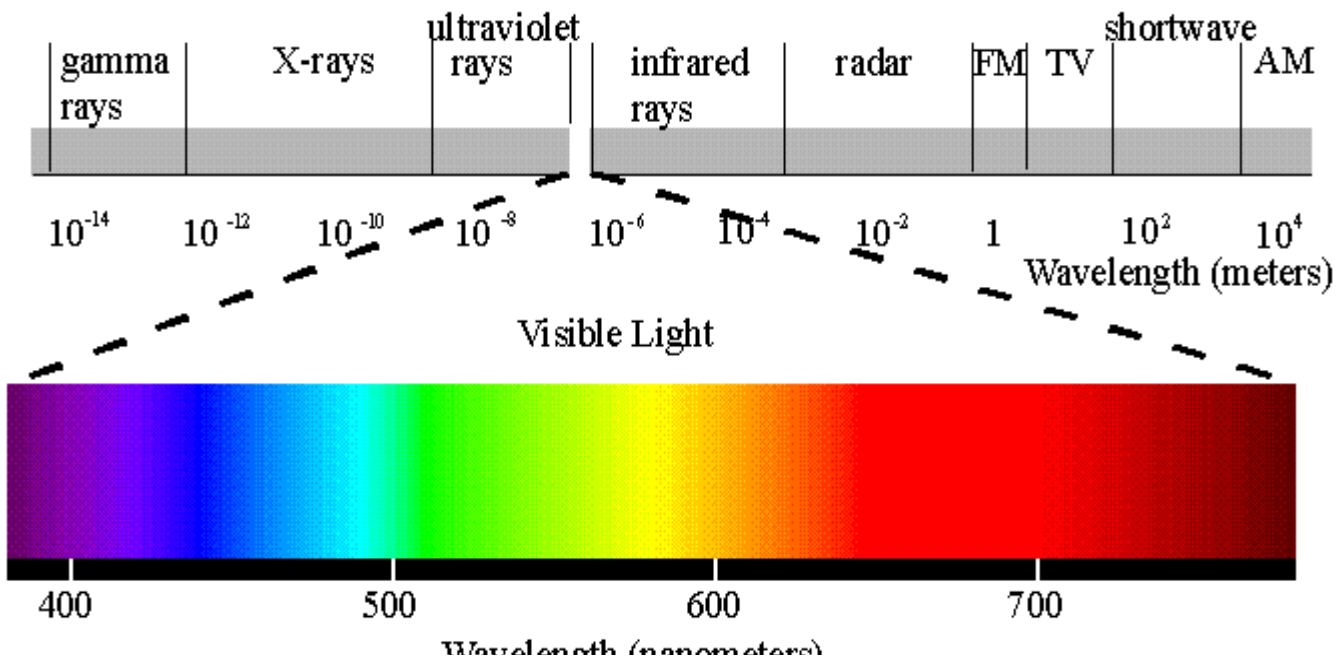
Such wave travels at the speed of light c . It does need a medium to propagate. It is made of a changing electric field E and of a changing magnetic field B . The fields are “shaking” As the wave propagates.

If you have AC current in an antenna it will create an E field That will induce a B field that will generate a E and so forth



<https://phet.colorado.edu/en/simulation/radio-waves>

Electromagnetic spectrum



Maxwell's Equations

Differential form

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

Maxwell's Equations

Integral form

$$\oint \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}$$

$$\oint \vec{E} \cdot d\vec{l} = -\int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{a}$$

$$\oint \vec{B} \cdot d\vec{a} = 0$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc} + \mu_0 \epsilon_0 \int \frac{\partial \vec{E}}{\partial t} \cdot d\vec{a}$$

8) equations of Maxwell = unification of magnetism and electricity

AND GOD SAID,"

$$\nabla \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial \tau} \longrightarrow \text{Faraday's law of induction. A changing magnetic field (H) induces an electric Field (E) or alternative current.}$$

$$\nabla \times \vec{H} = \vec{J}_c + \epsilon \frac{\partial \vec{E}}{\partial \tau} \longrightarrow \text{Ampere's law completed by Maxwell. A direct current (J) creates a magnetic field. A changing electric field. creates a changing magnetic field. Even in space, in Absence of wire. That's what makes light. Auto generating E and B.}$$

$$\nabla \cdot \vec{D} = \rho_v \longrightarrow \text{Gauss's law. A distribution of charge creates an electric field in space. The lines Don't make close loop. The lines diverge from a source and converge to a sink (Van Graaf)}$$

$$\nabla \cdot \vec{B} = 0 \longrightarrow \text{Magnetic field make loop. They leave the north pole and come back into the south pole Back into the North. You can't single out a pole . You can single out a charge.}$$

LET THERE BE
LIGHT."

Playing with the equations, Maxwell found the speed of light C ($3 \cdot 10^8$ m/s).

When we compute the electric field, and the electric force between charges, we include a parameter $K_e = 9 \cdot 10^9$
For example: electric field from a point charge $= E = k_e q / d^2$

When we compute the magnetic field we include a parameter $K_m = 10^{-7}$
magnetic field produced by a wire $= B = K_m 2I/d$
if you compute the ratio K_e/K_m amazingly you get
 $9 \cdot 10^{16} \text{ (m/s)}^2$ which is the square of the speed of light !!!!